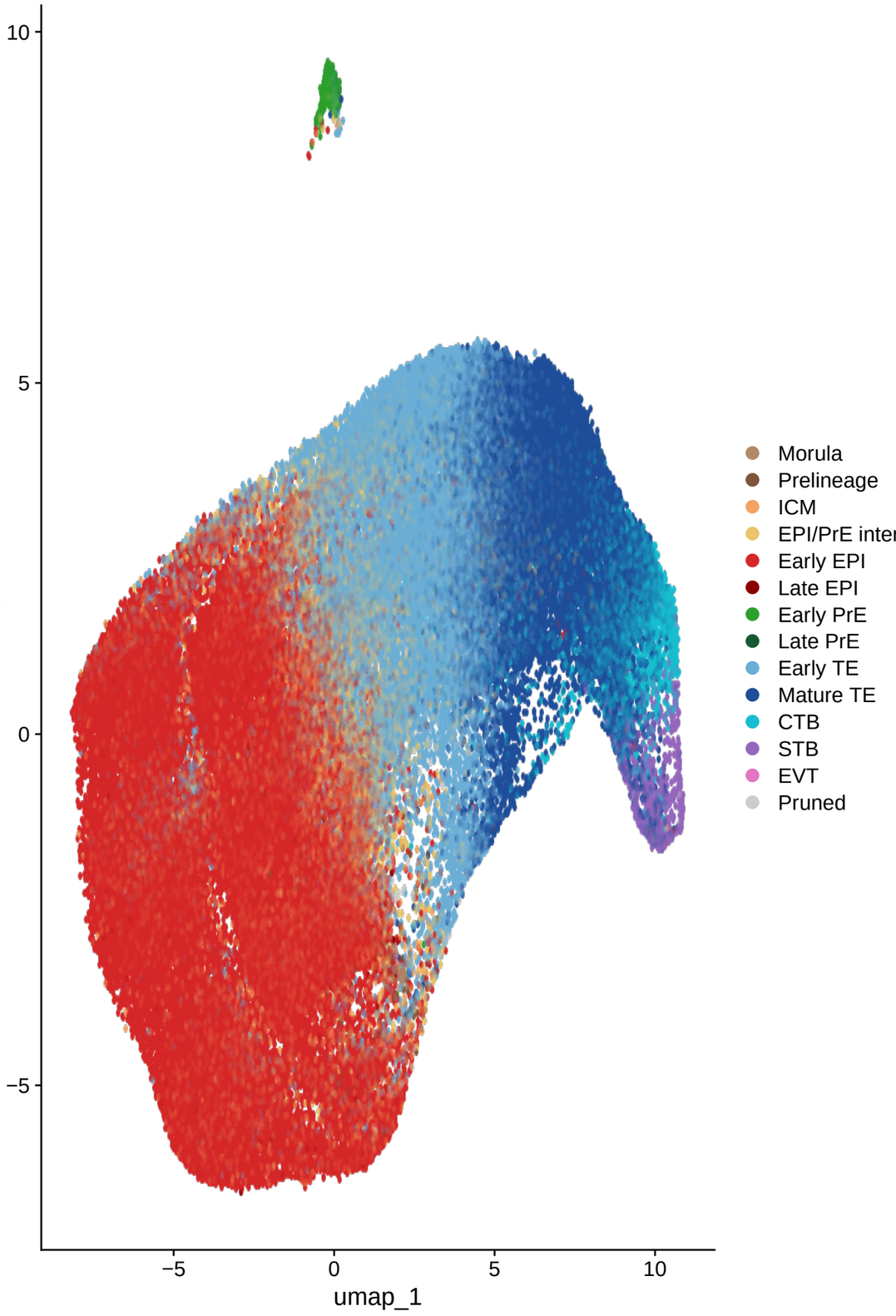
Reference labels by stage (2173 cells; CS7 gastrula removed, amnion kept)

	Zygote	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E12	E14	CS7
Cleavage	2	6	12	84										
Morula						190	12							
Prelineage							129							
ICM							51							
EPI/PrE intermediate							8	5	3					
Early EPI							23	103	50	9	2	1		1
Late EPI										1	8	12	18	31
Early PrE							2	29	39	2	3	2		
Late PrE												1	8	6
Early TE						110	407							
Mature TE								449	16					
CTB							1	2	17	42	30	24	5	
STB									13	13	29	27	30	
EVT												4	37	
ExE mesoderm												2	14	
Amnion														48

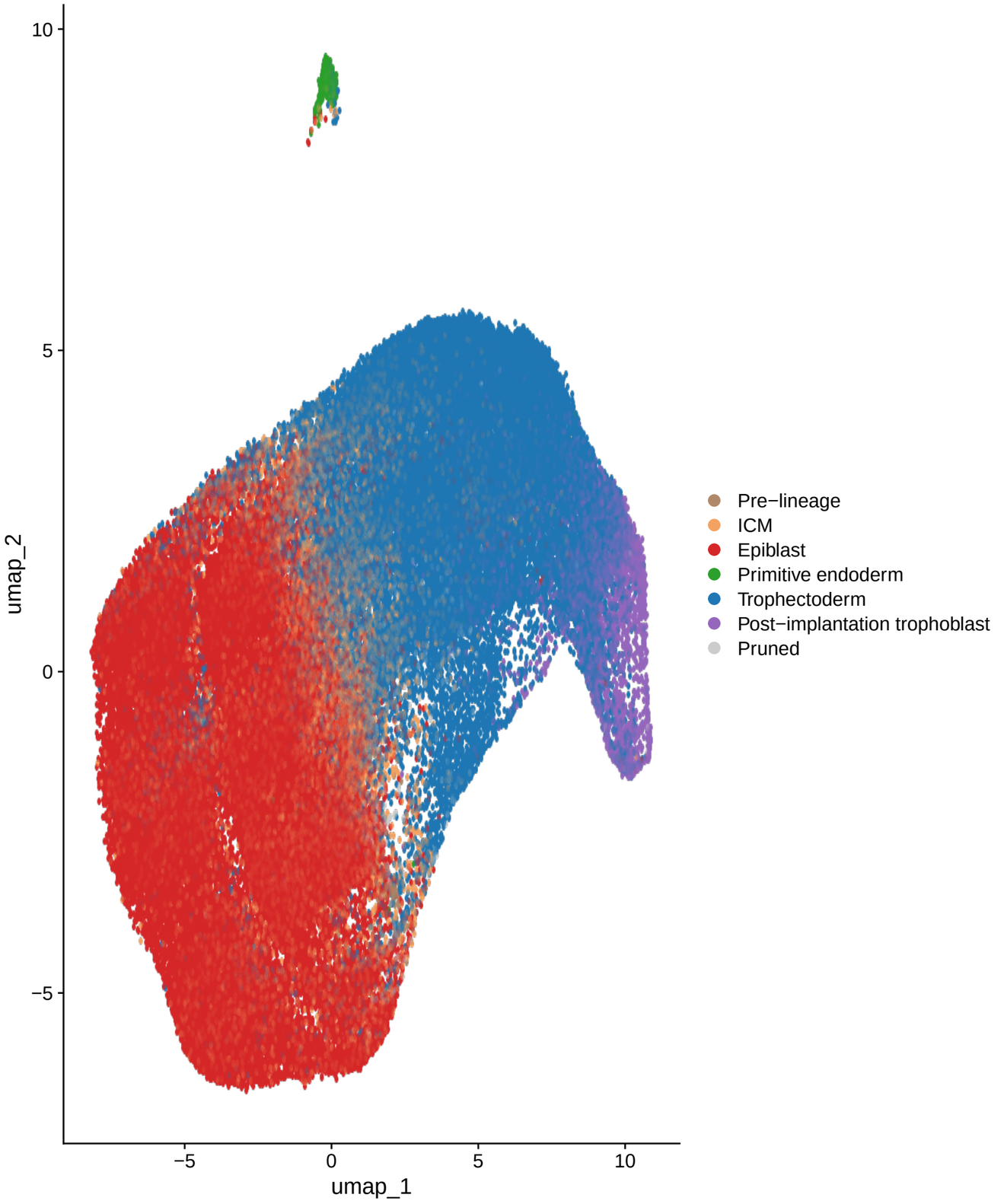
cells: 400, 300, 200, 100

Stage

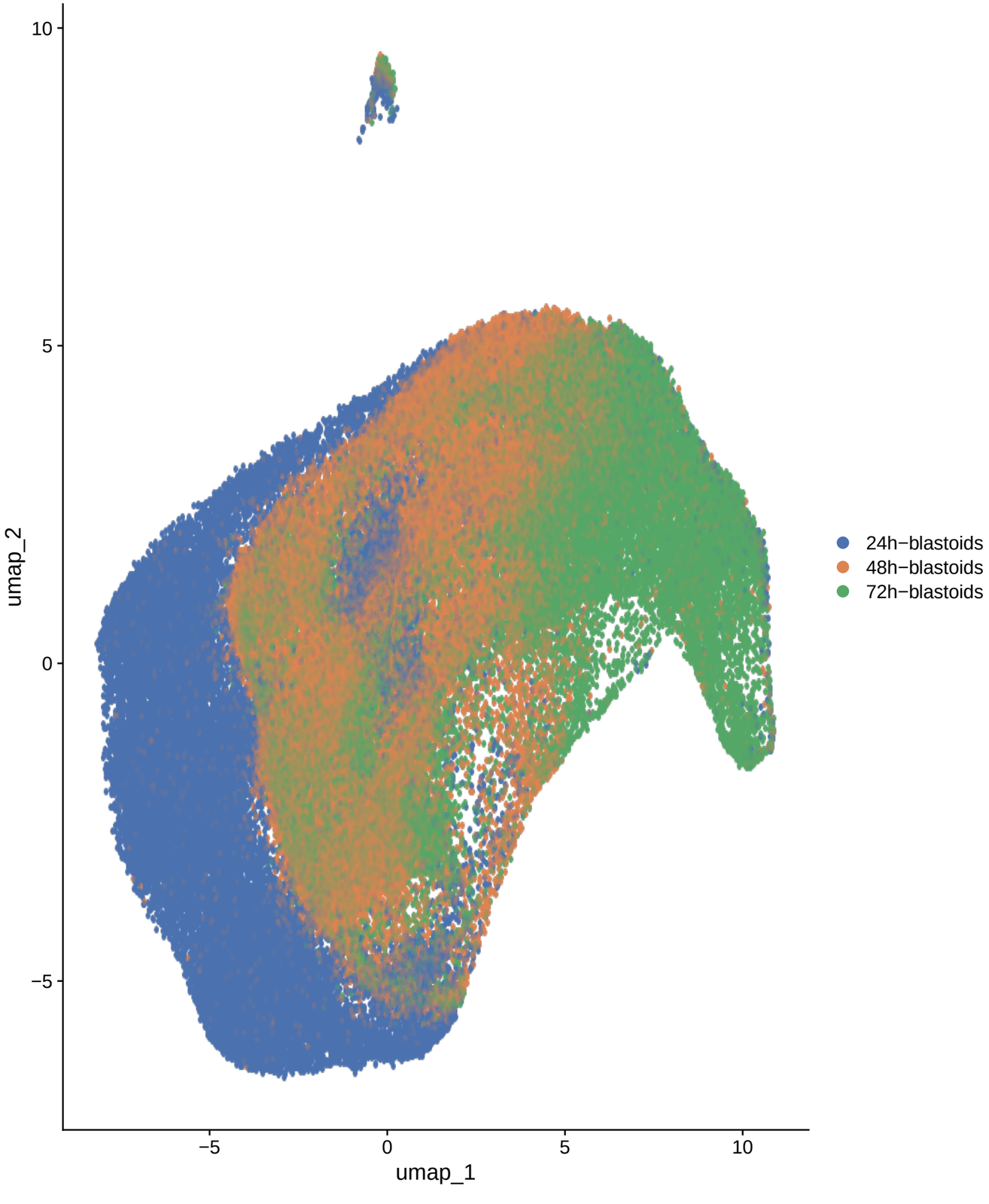
Per−cell SingleR label (grey = pruned)



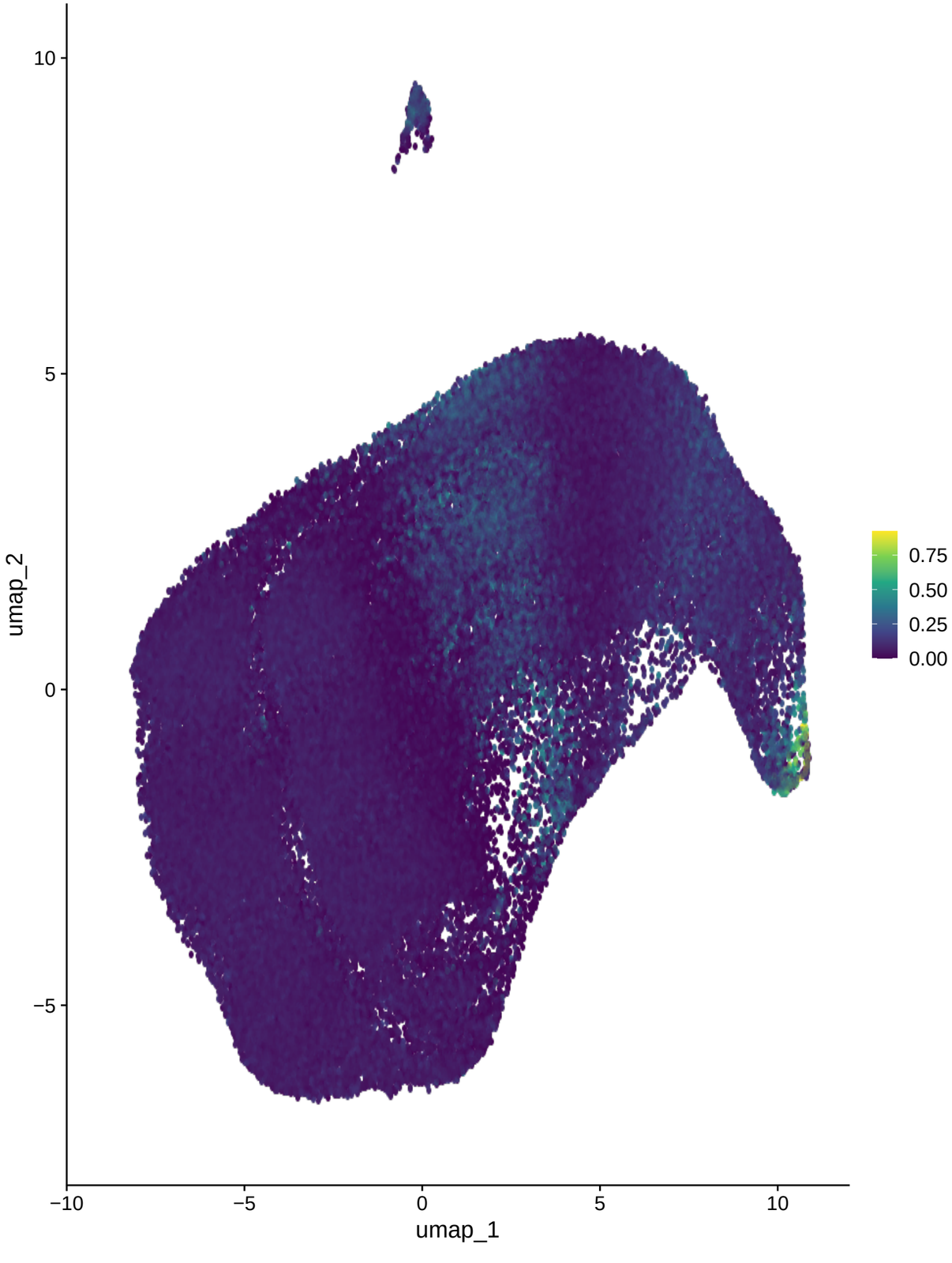
Lineage



Sample



Delta to the next label (low = ambiguous)



Mean SingleR score (Spearman) of the cells given each label, against every reference label

Assigned label (cells)	Cleavage	Morula	Prelineage	ICM	EPI/PrE intermediate	Early EPI	Late EPI	Early PrE	Late PrE	Early TE	Mature TE	CTB	STB	EVT	ExE mesoderm	Amnion
Morula (67)	0.36	0.45	0.45	0.44	0.41	0.42	0.38	0.38	0.30	0.40	0.37	0.32	0.28	0.23	0.28	0.21
Prelineage (431)	0.21	0.34	0.40	0.40	0.40	0.40	0.36	0.37	0.30	0.39	0.36	0.32	0.27	0.24	0.26	0.20
ICM (3615)	0.20	0.32	0.37	0.40	0.39	0.41	0.37	0.36	0.30	0.37	0.34	0.31	0.27	0.23	0.26	0.20
EPI/PrE intermediate (4008)	0.20	0.33	0.41	0.42	0.44	0.43	0.38	0.40	0.33	0.43	0.40	0.36	0.31	0.27	0.28	0.22
Early EPI (47735)	0.20	0.32	0.38	0.41	0.40	0.42	0.40	0.37	0.31	0.37	0.35	0.32	0.27	0.24	0.28	0.21
Late EPI (143)	0.18	0.28	0.32	0.35	0.34	0.37	0.37	0.32	0.29	0.33	0.31	0.30	0.26	0.24	0.27	0.20
Early PrE (464)	0.17	0.29	0.36	0.40	0.43	0.39	0.32	0.47	0.41	0.40	0.39	0.35	0.30	0.28	0.31	0.21
Late PrE (1)	0.14	0.21	0.27	0.30	0.32	0.30	0.25	0.35	0.35	0.32	0.31	0.30	0.25	0.25	0.26	0.17
Early TE (19473)	0.20	0.33	0.42	0.41	0.45	0.43	0.36	0.42	0.33	0.46	0.44	0.40	0.34	0.30	0.29	0.24
Mature TE (14181)	0.18	0.28	0.36	0.35	0.43	0.40	0.33	0.43	0.33	0.46	0.47	0.45	0.39	0.35	0.30	0.24
CTB (2321)	0.16	0.23	0.31	0.30	0.38	0.36	0.31	0.41	0.32	0.42	0.45	0.48	0.42	0.39	0.28	0.22
STB (744)	0.12	0.18	0.24	0.24	0.32	0.30	0.26	0.37	0.28	0.35	0.42	0.46	0.48	0.38	0.22	0.17
EVT (2)	0.12	0.19	0.25	0.25	0.31	0.31	0.31	0.38	0.33	0.36	0.42	0.56	0.51	0.58	0.30	0.22

Reference label

mean score
0.5
0.4
0.3
0.2

Delta to the next label per cell (0.4% pruned)

delta

0.75

0.50

0.25

0.00

Morula

Prelineage

ICM

EPI/PrE intermediate

Early EPI

Late EPI

Early PrE

Late PrE

Early TE

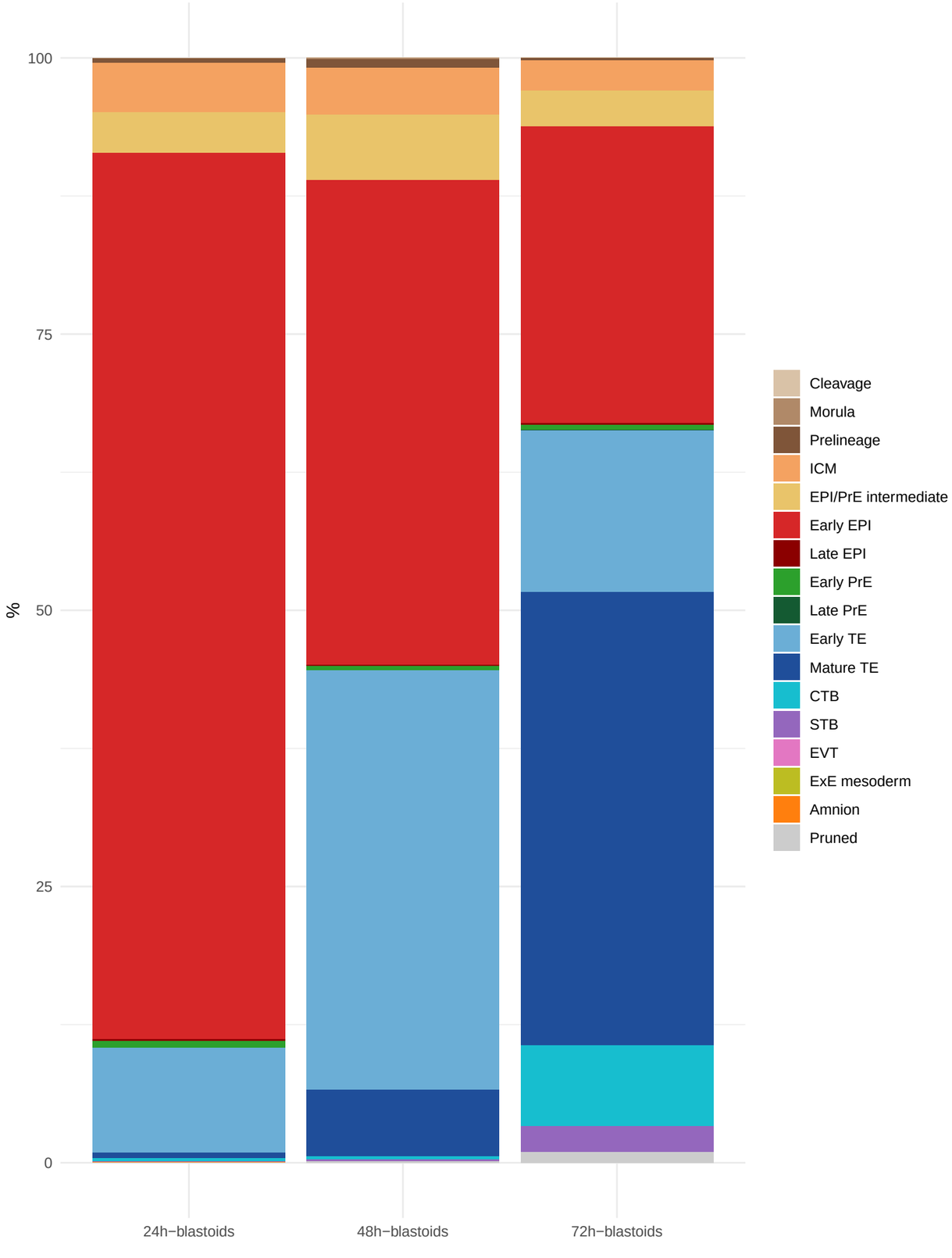
Mature TE

CTB

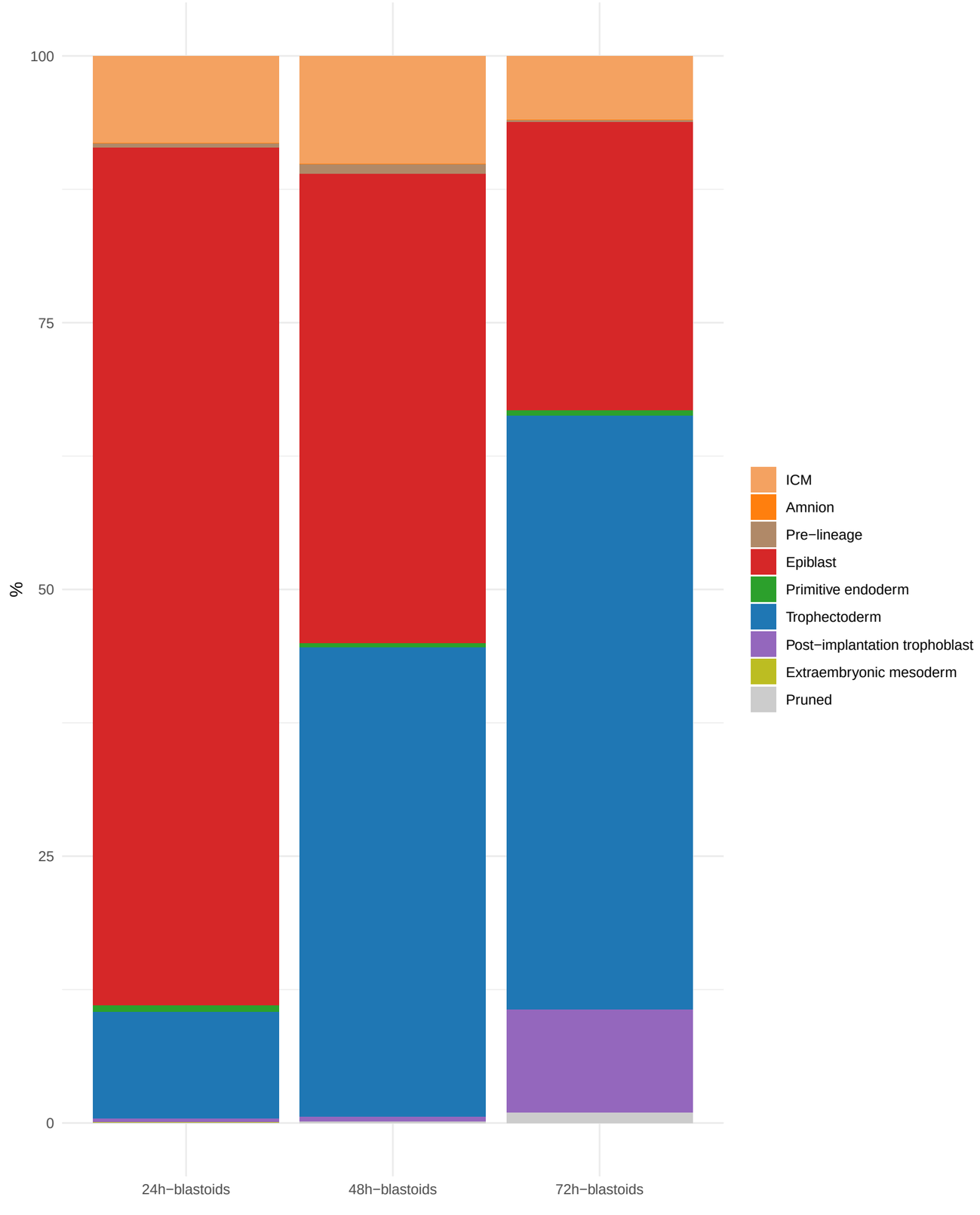
STB

EVT

Per−cell labels by sample

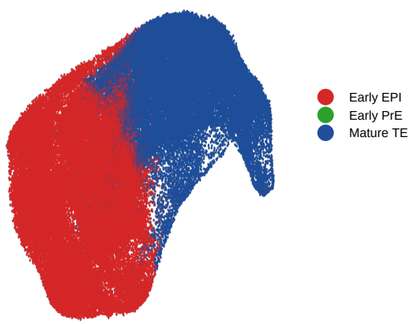


Per−cell lineages by sample

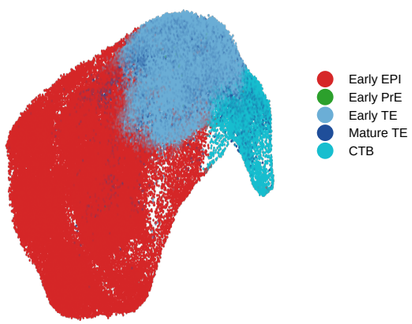


SingleR cluster labels at each resolution

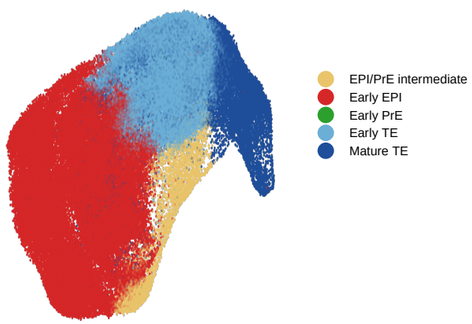
res 0.1 (4 clusters)



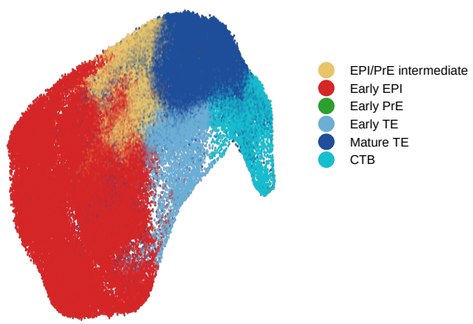
res 0.2 (7 clusters)



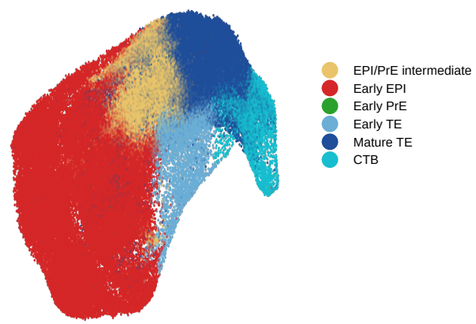
res 0.3 (11 clusters)



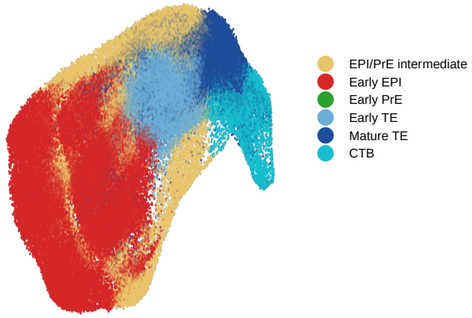
res 0.4 (12 clusters)



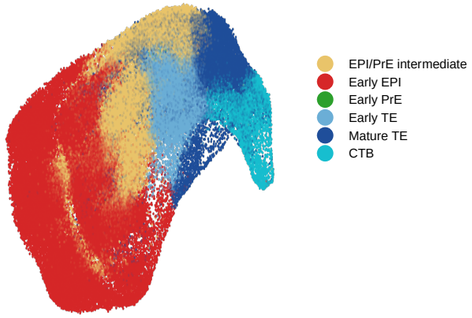
res 0.5 (14 clusters)



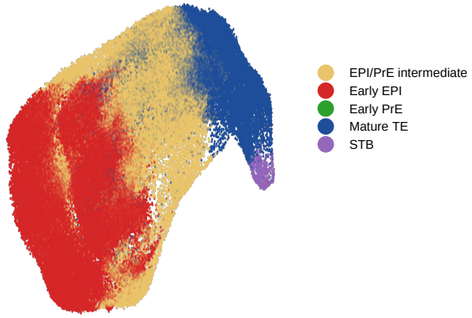
res 0.6 (16 clusters)



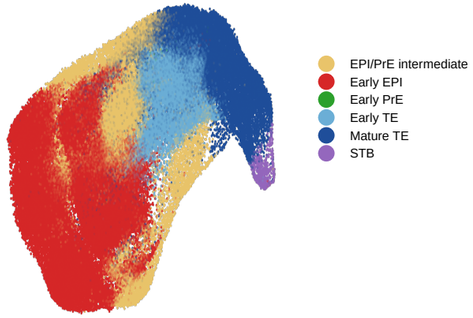
res 0.7 (19 clusters)



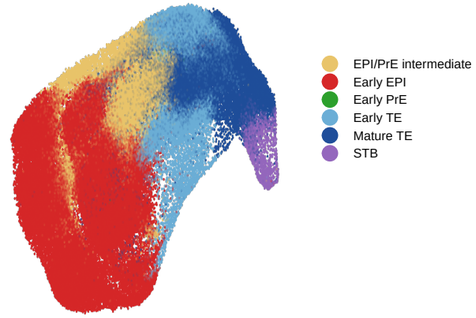
res 0.8 (20 clusters)



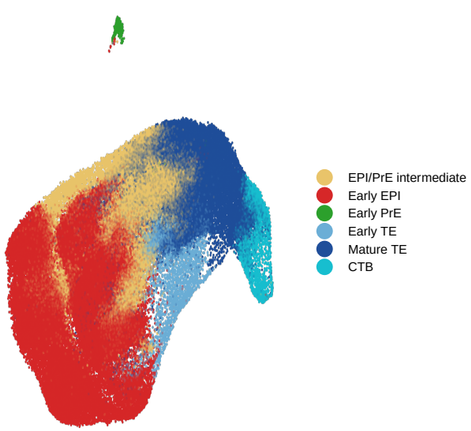
res 0.9 (22 clusters)



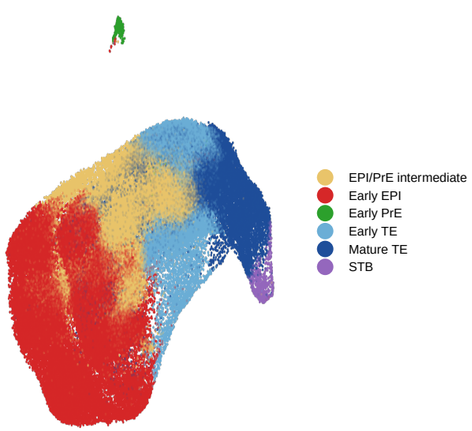
res 1 (23 clusters)



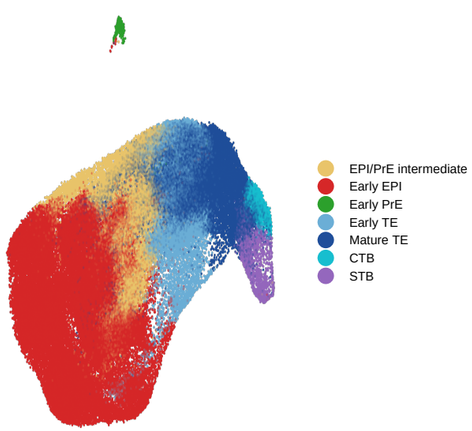
res 1.1 (24 clusters)



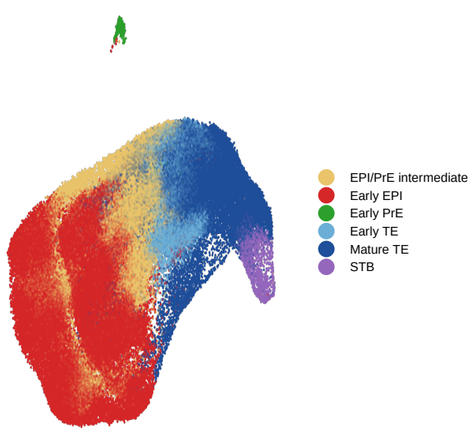
res 1.2 (24 clusters)



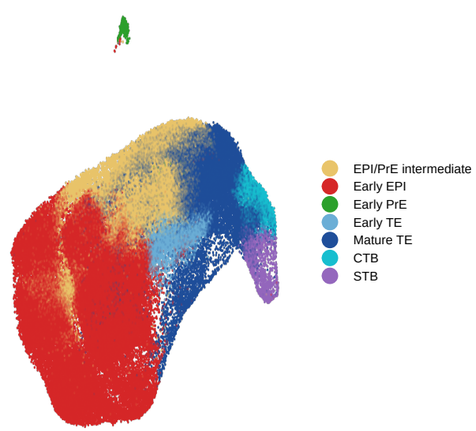
res 1.3 (27 clusters)



res 1.4 (28 clusters)

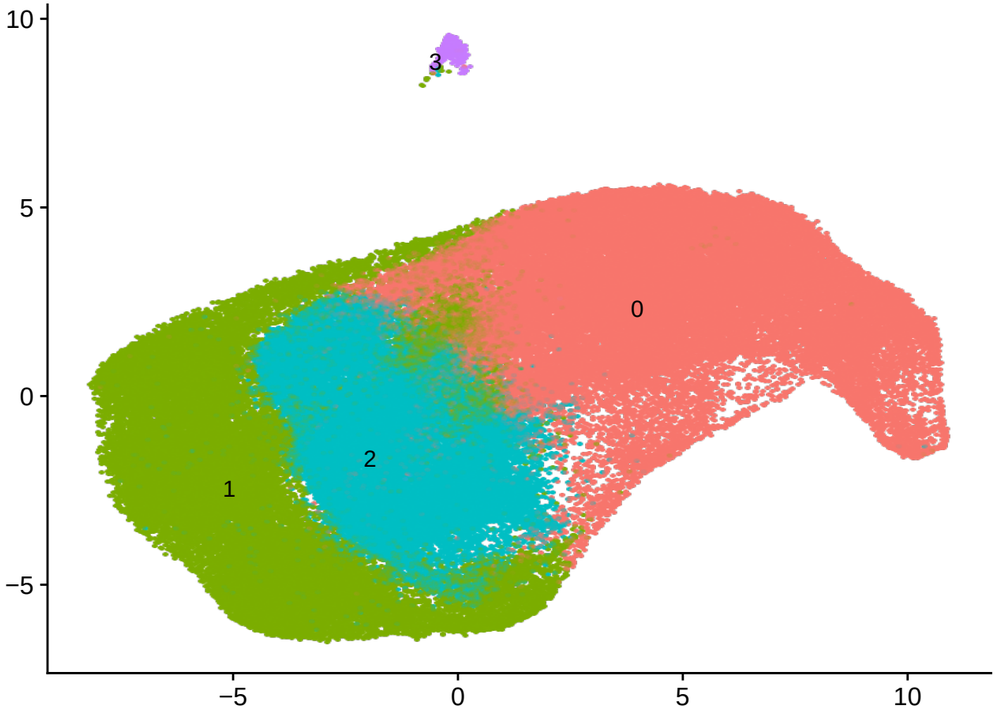


res 1.5 (28 clusters)

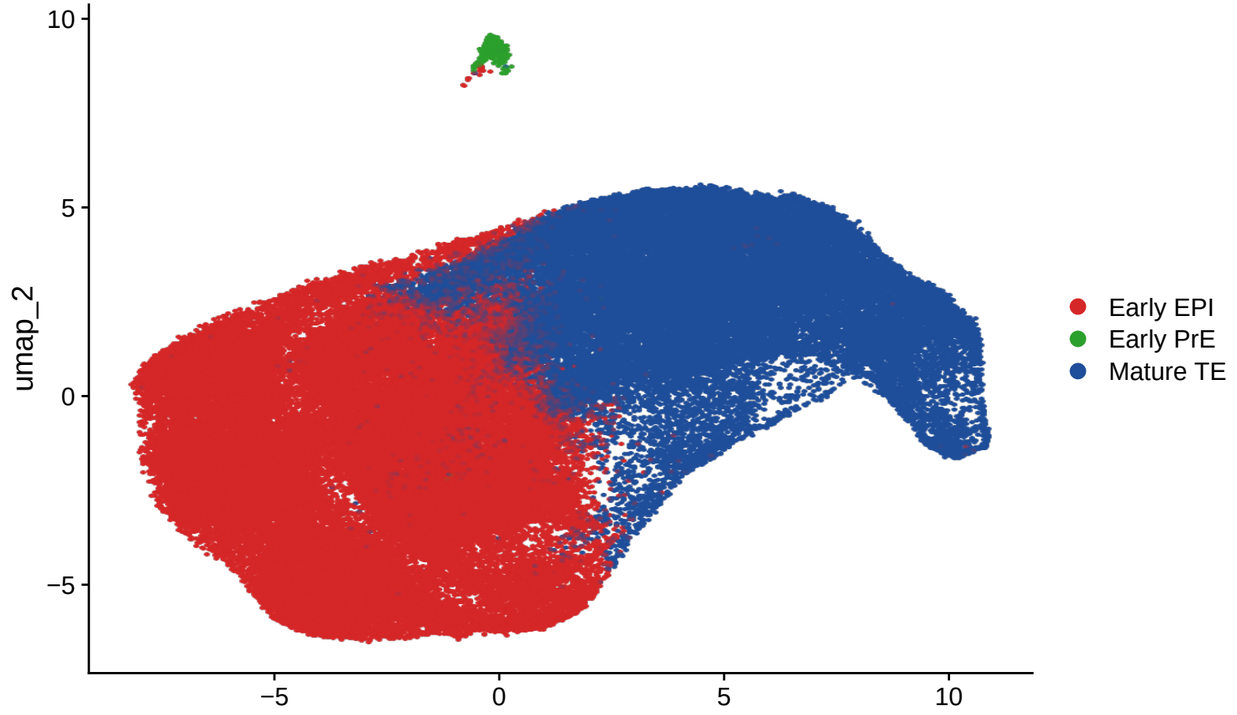


Resolution 0.1

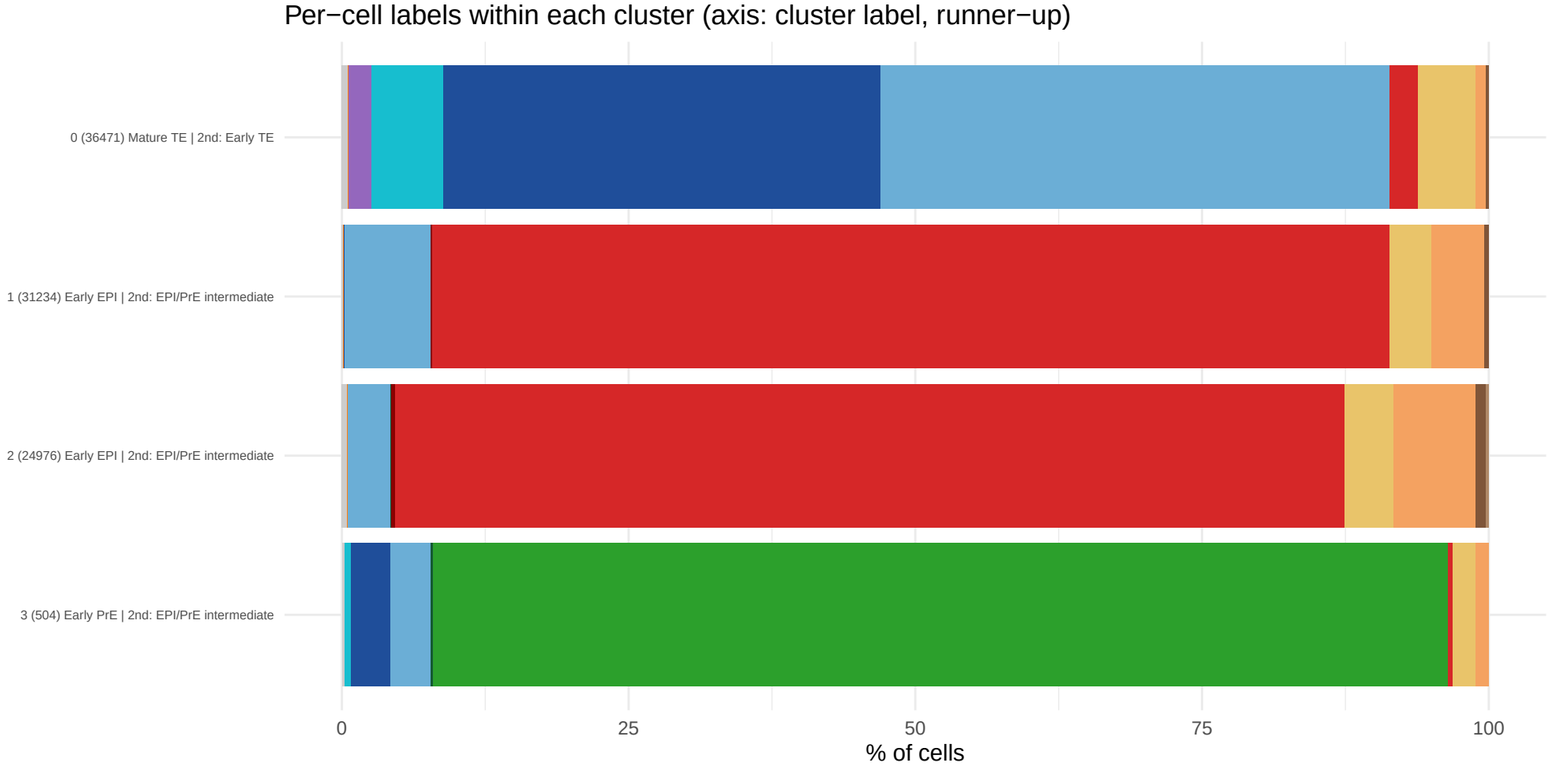
Clusters (4)



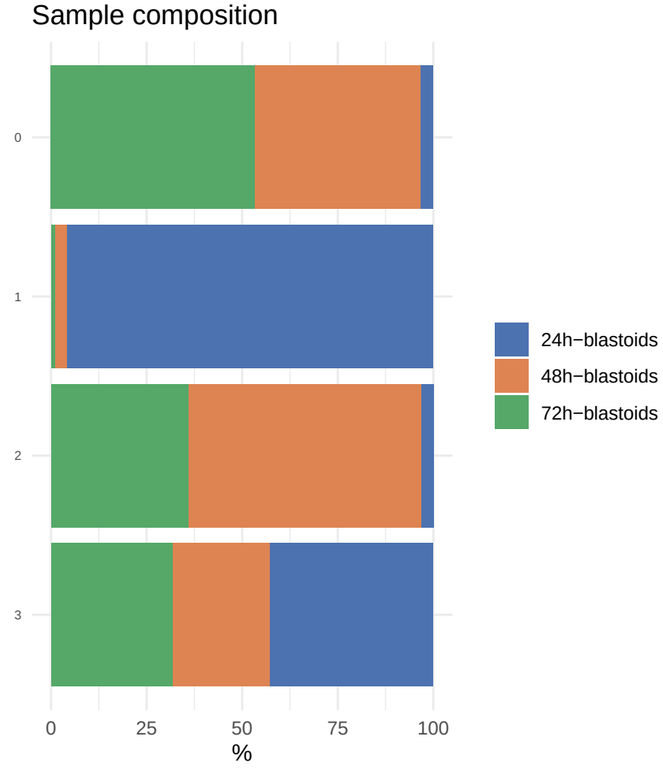
SingleR cluster label



Per−cell labels within each cluster (axis: cluster label, runner−up)

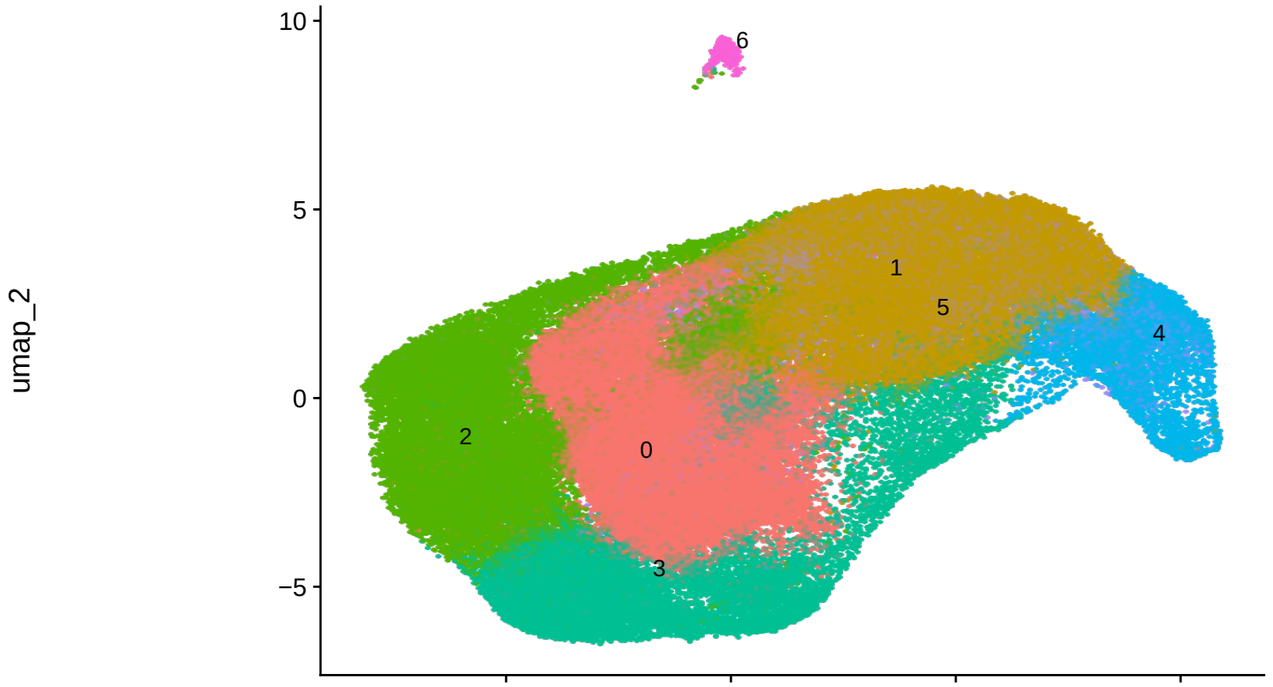


Sample composition

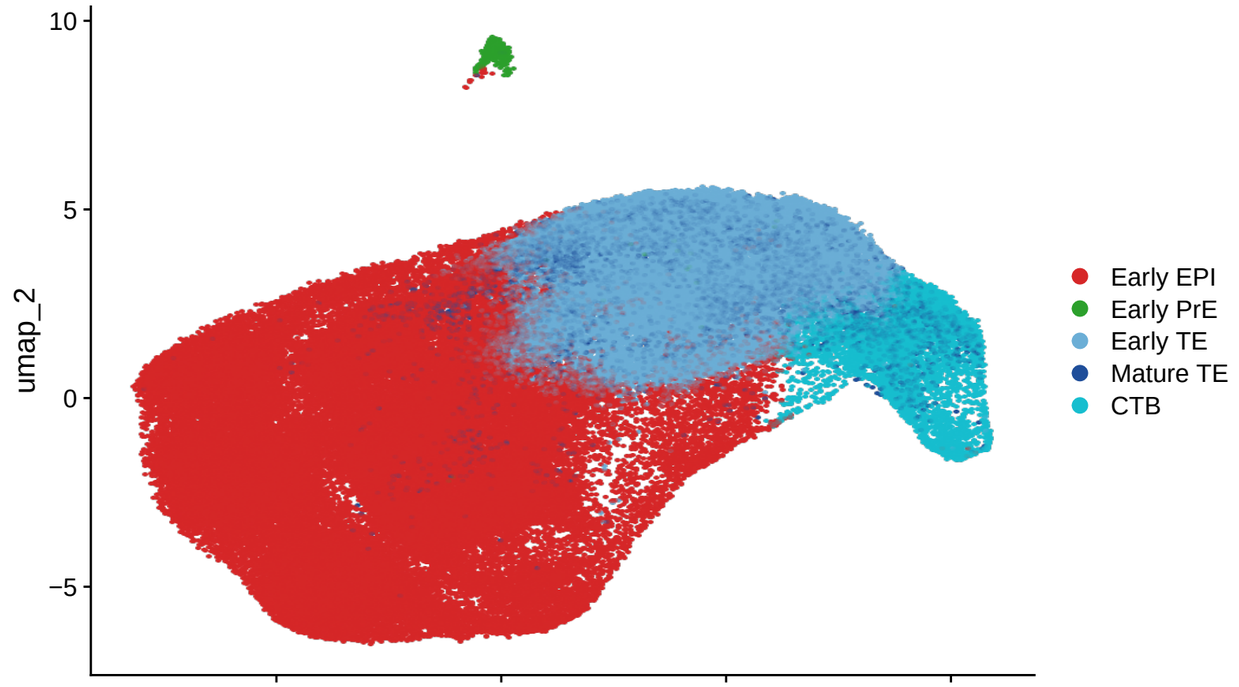


Resolution 0.2

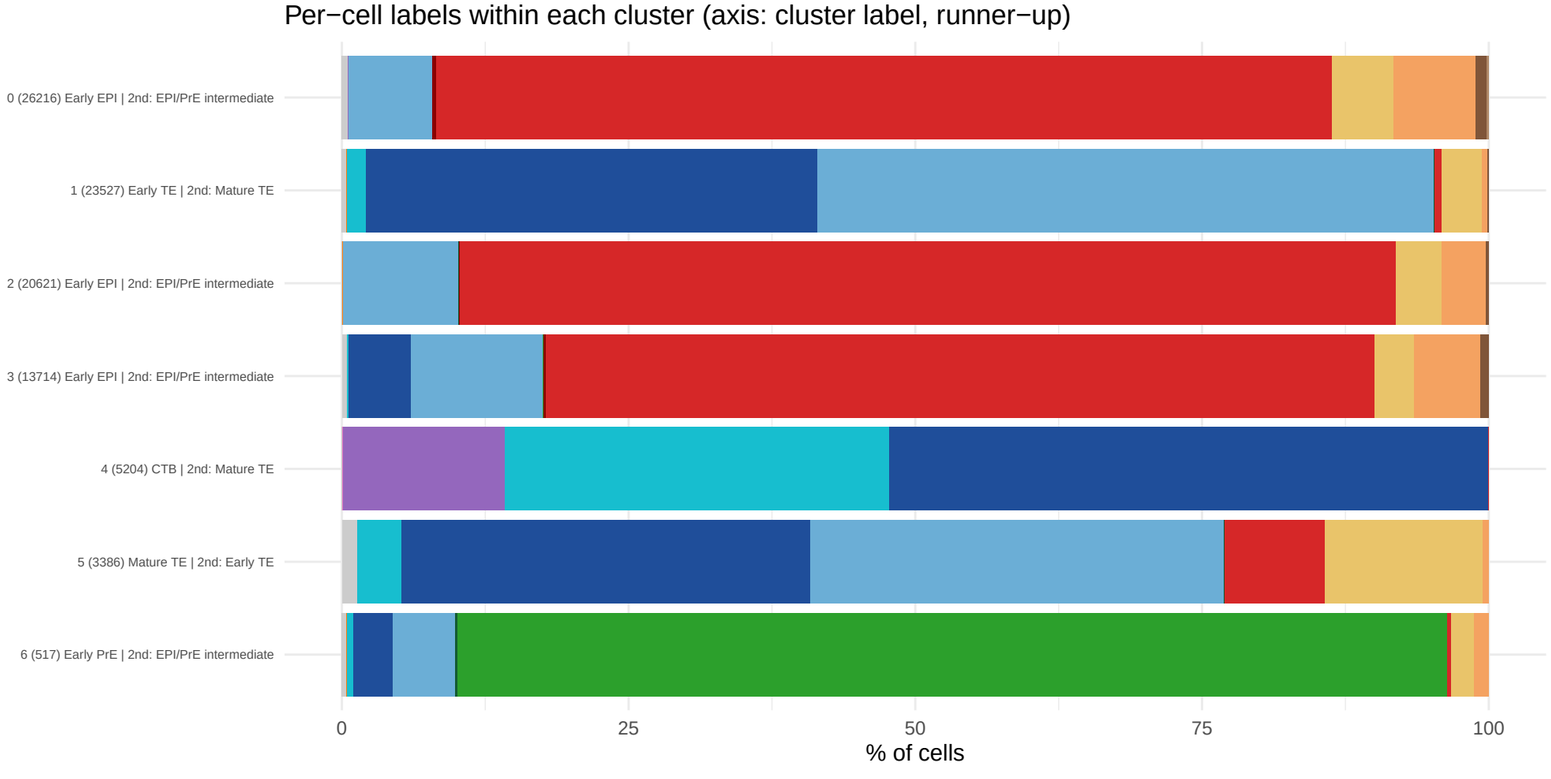
Clusters (7)



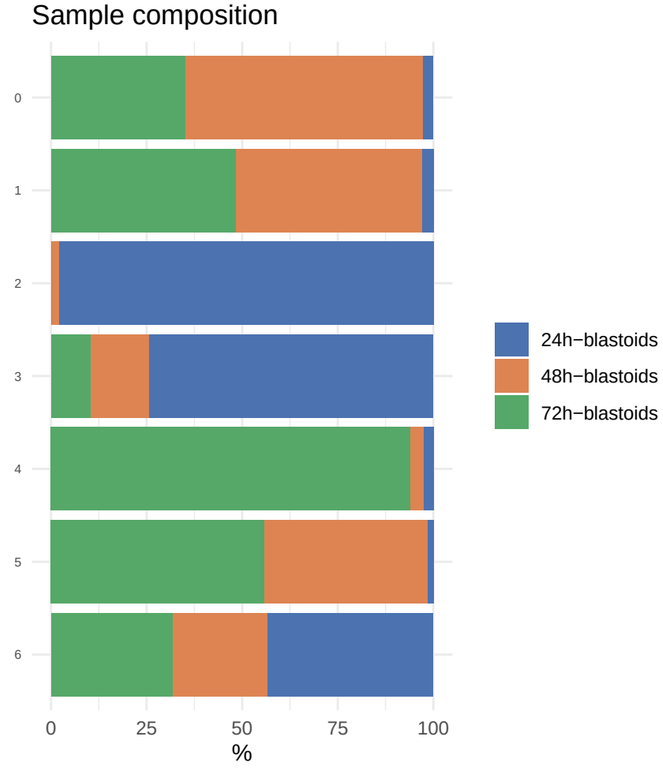
SingleR cluster label



Per−cell labels within each cluster (axis: cluster label, runner−up)

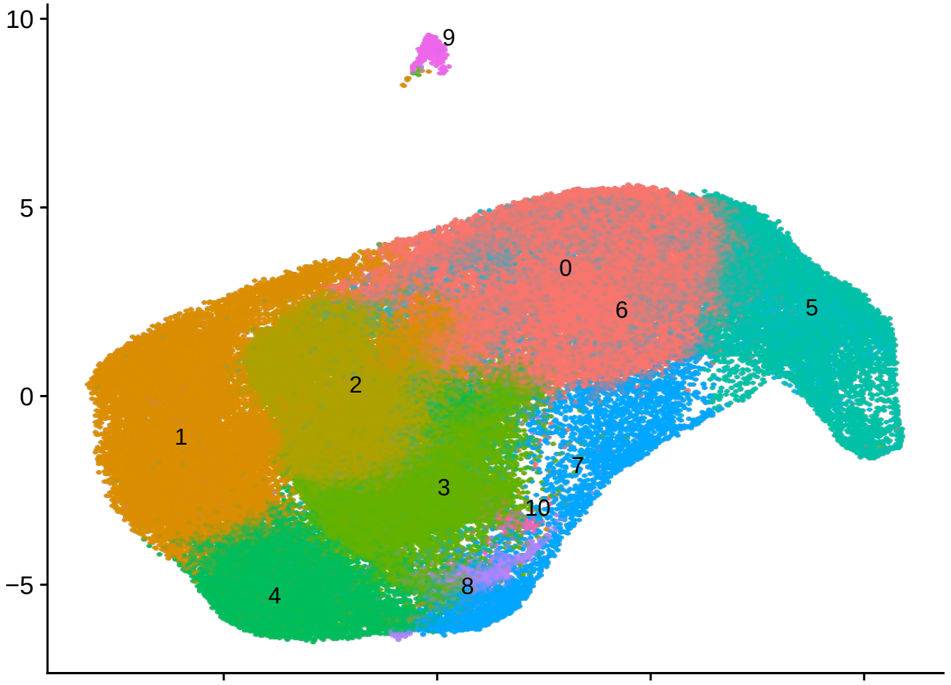


Sample composition

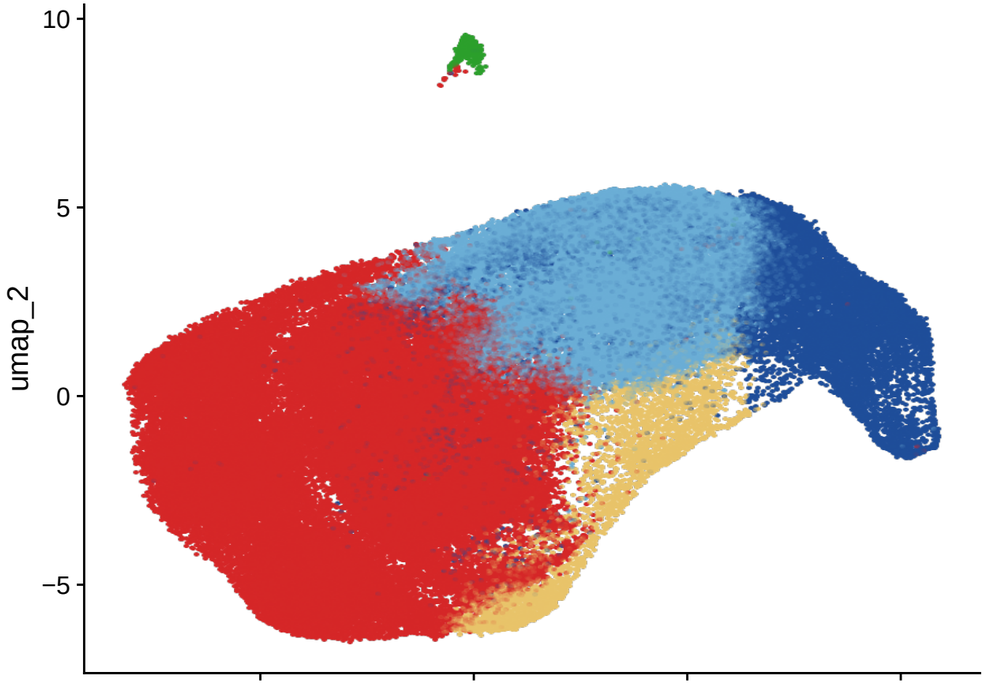


Resolution 0.3

Clusters (11)

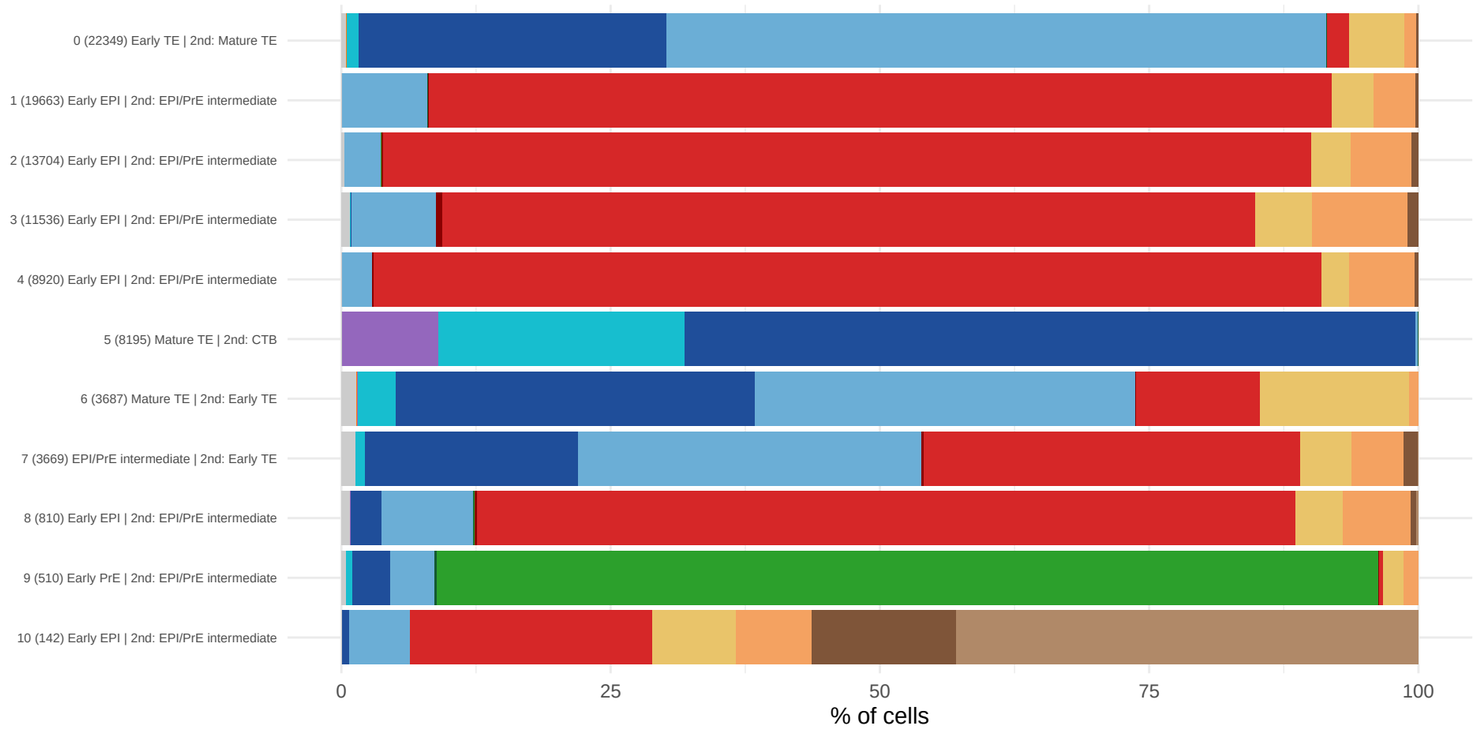


SingleR cluster label



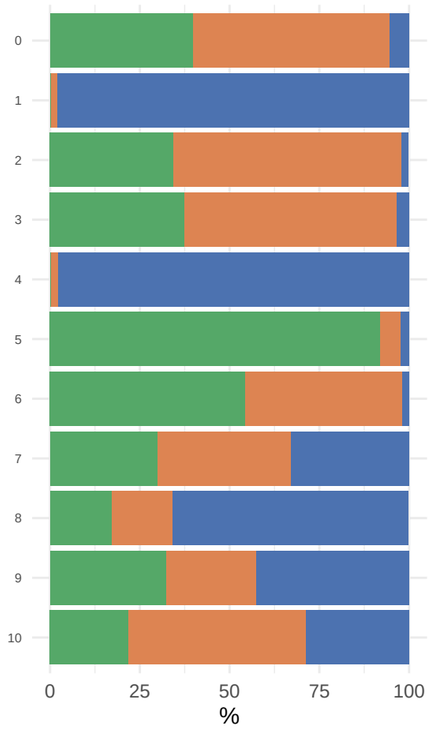
EPI/PrE intermediate
Early EPI
Early PrE
Early TE
Mature TE

Per−cell labels within each cluster (axis: cluster label, runner−up)



Cleavage
Morula
Prelineage
ICM
EPI/PrE intermediate
Early EPI
Late EPI
Early PrE
Late PrE
Early TE
Mature TE
CTB
STB
EVT
ExE mesoderm
Amnion
Pruned

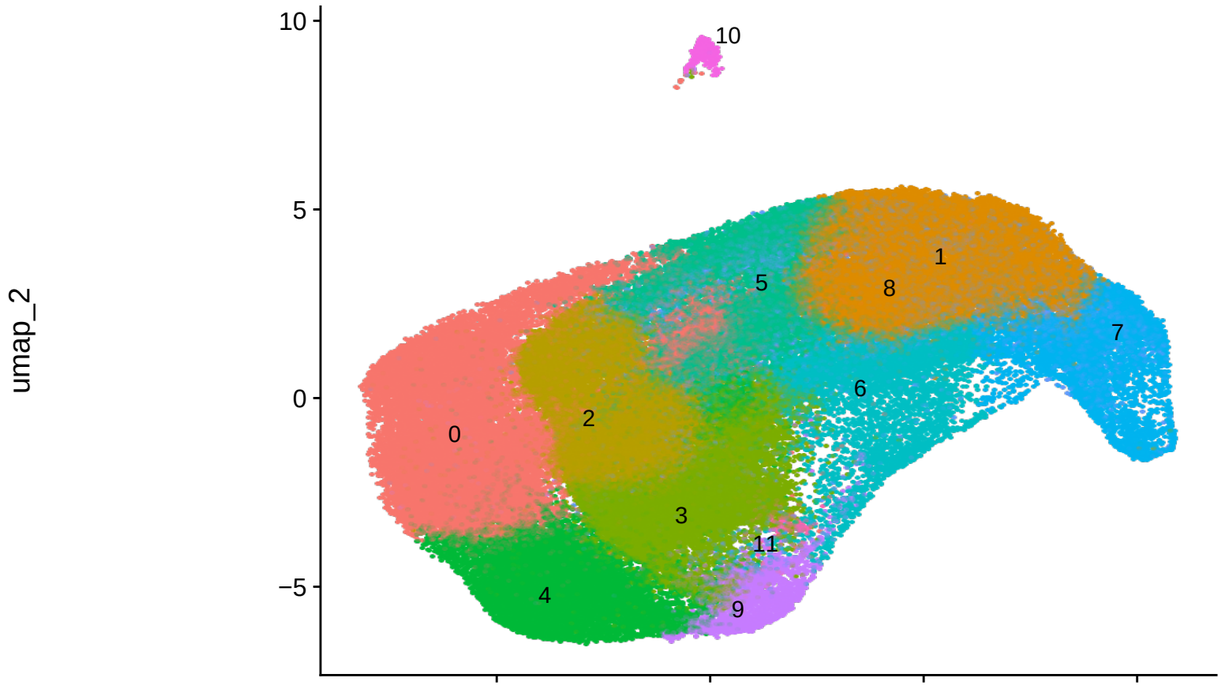
Sample composition



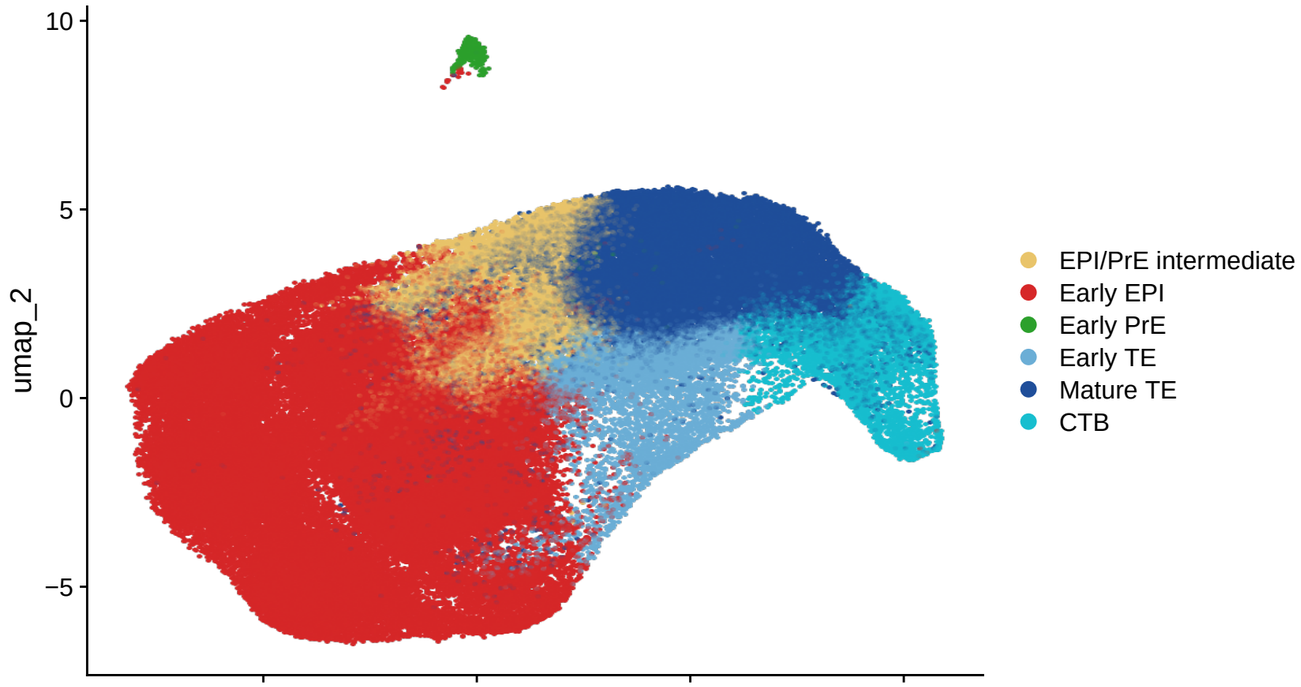
24h−blastoids
48h−blastoids
72h−blastoids

Resolution 0.4

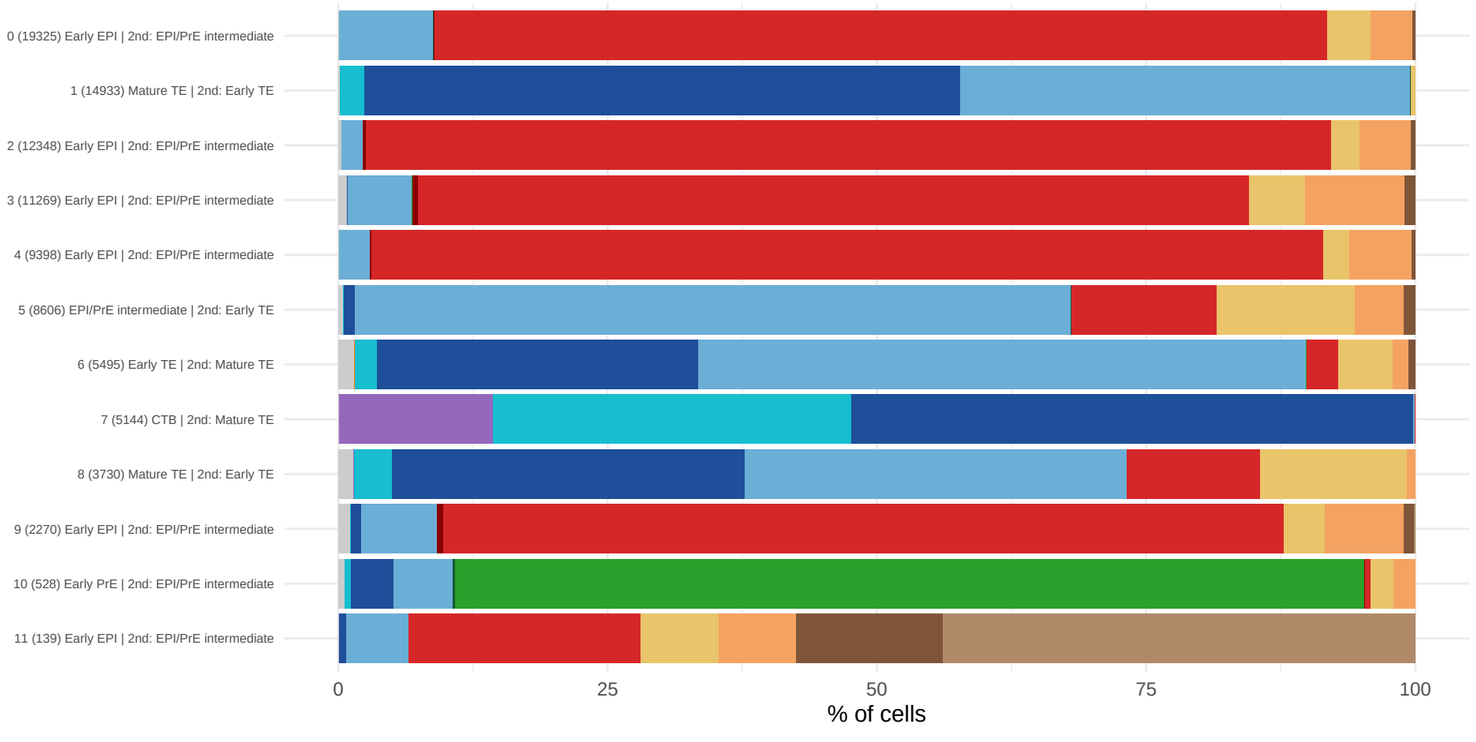
Clusters (12)



SingleR cluster label

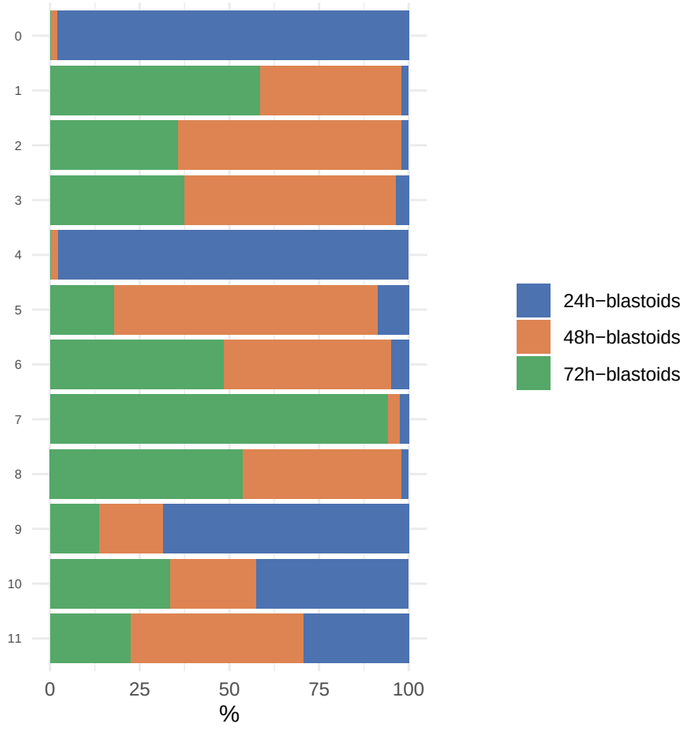


Per−cell labels within each cluster (axis: cluster label, runner−up)



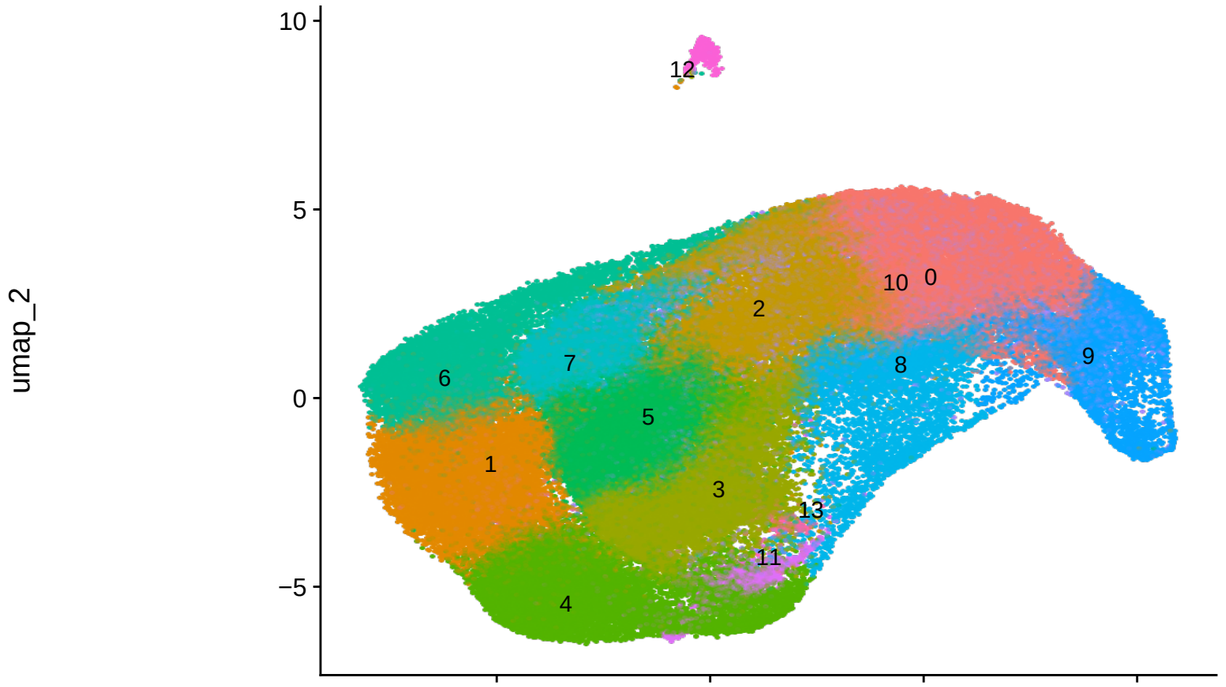


Sample composition

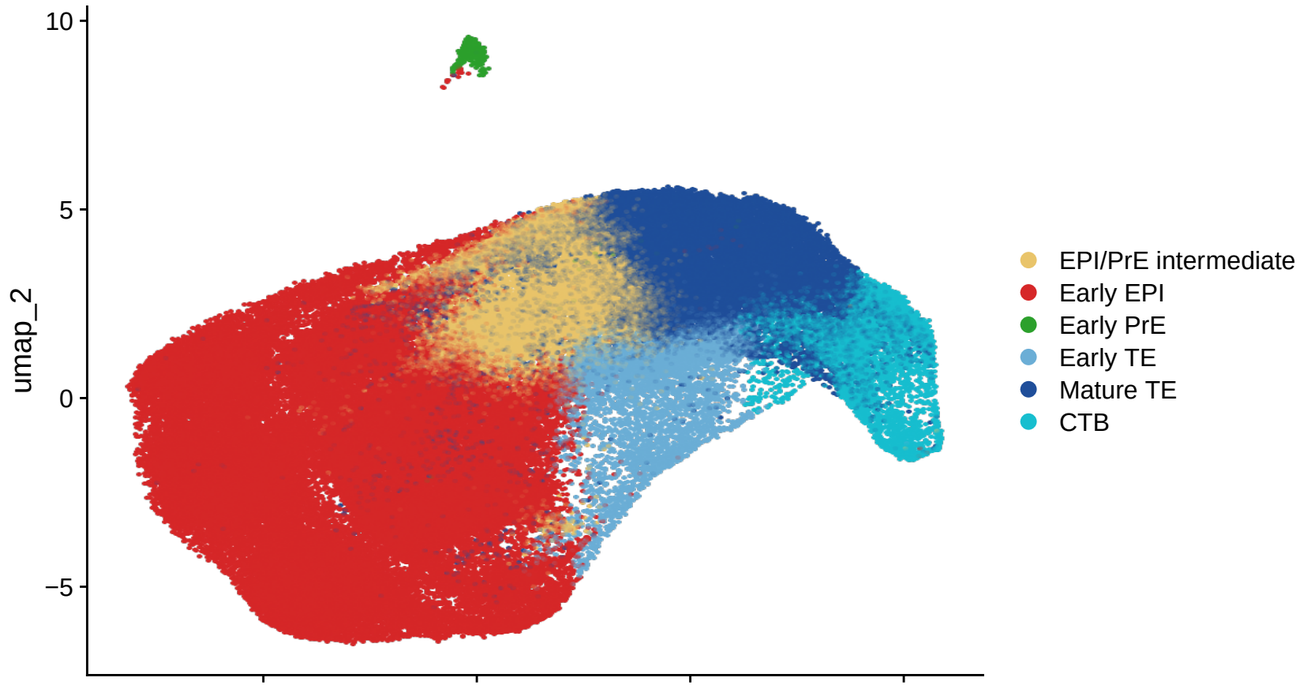


Resolution 0.5

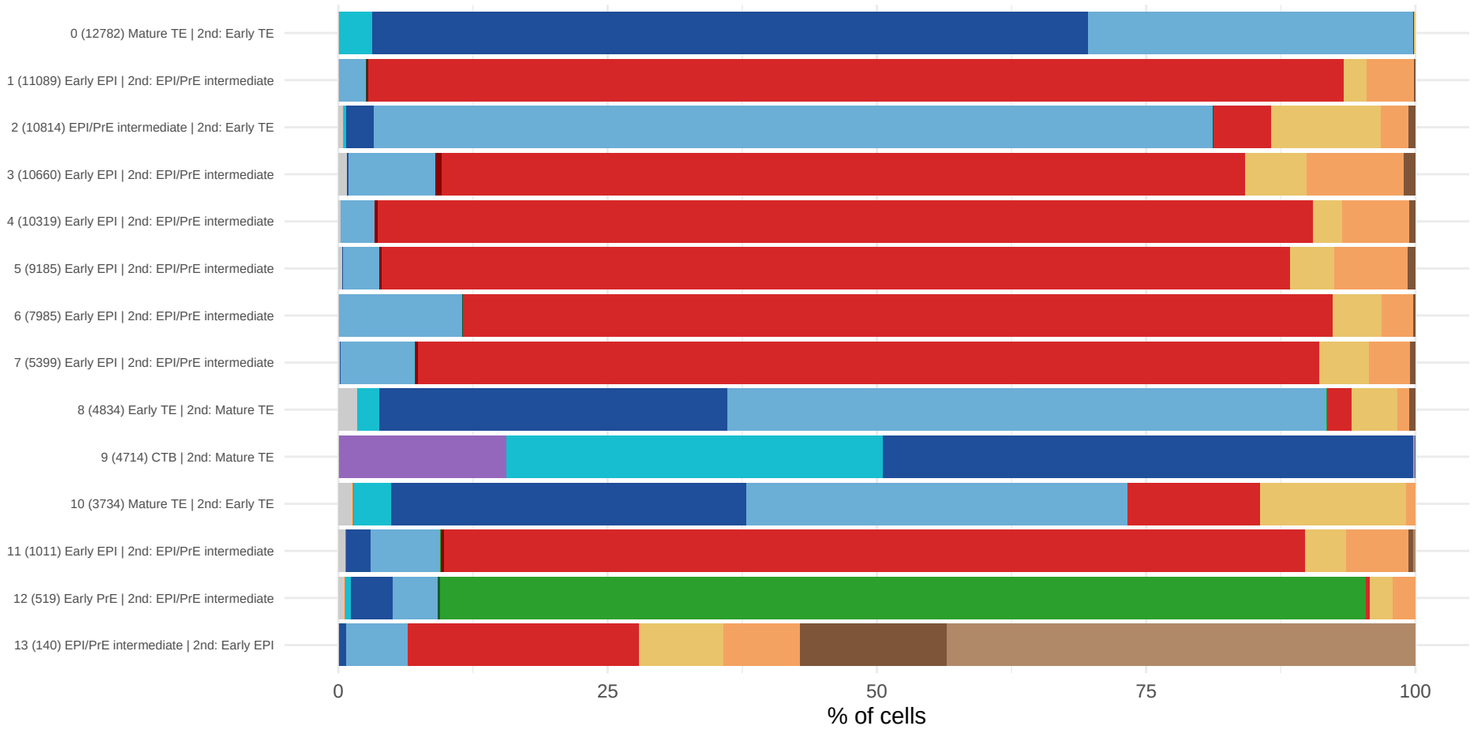
Clusters (14)



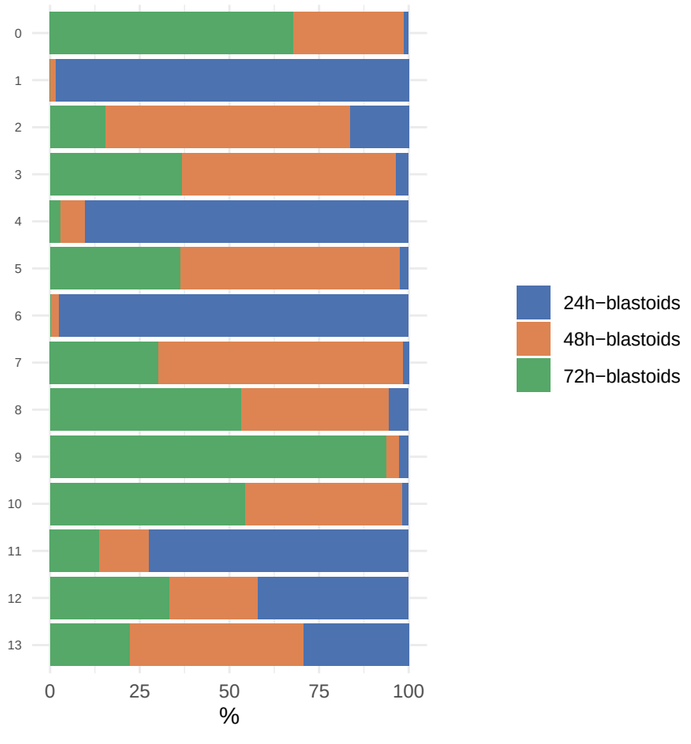
SingleR cluster label



Per−cell labels within each cluster (axis: cluster label, runner−up)

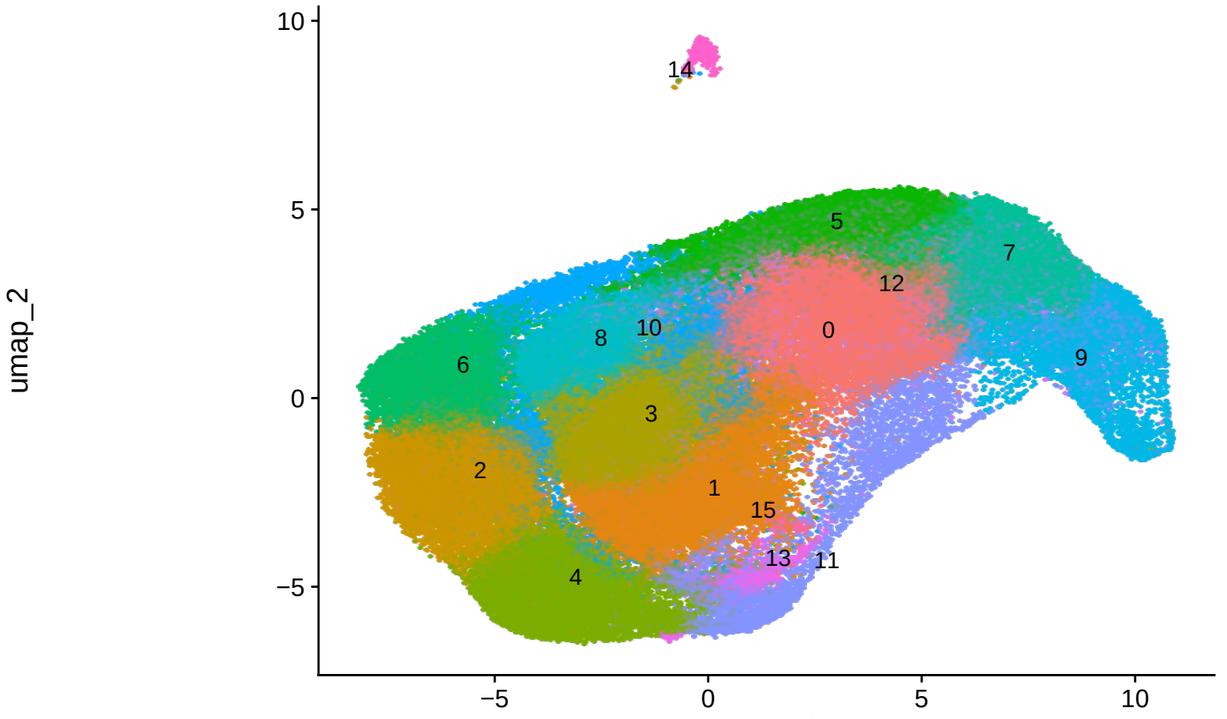


Sample composition

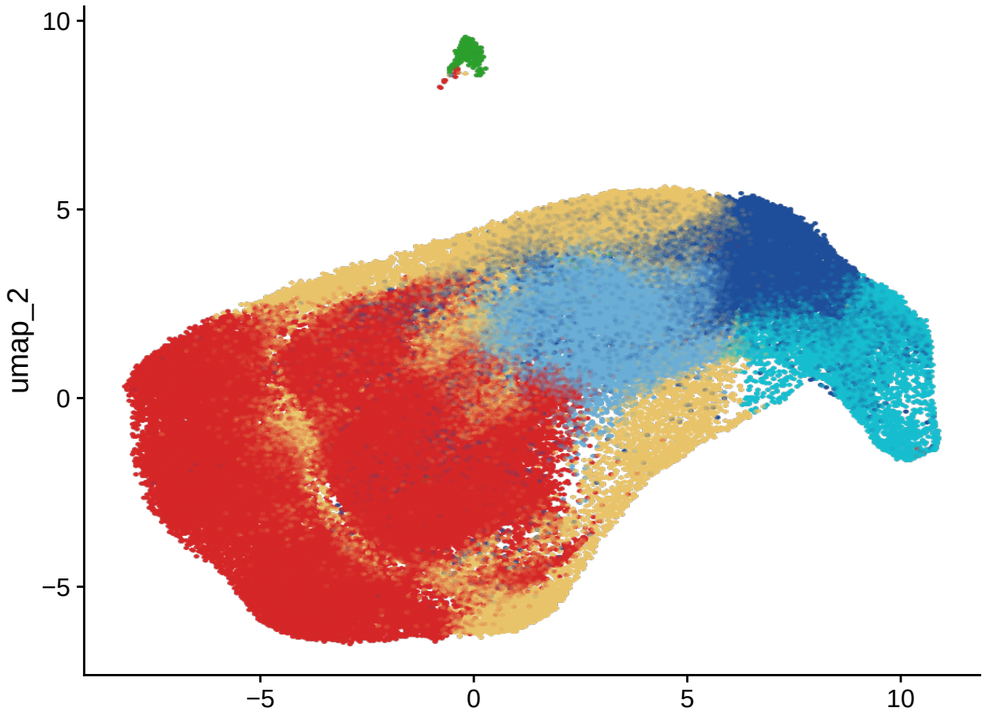


Resolution 0.6

Clusters (16)

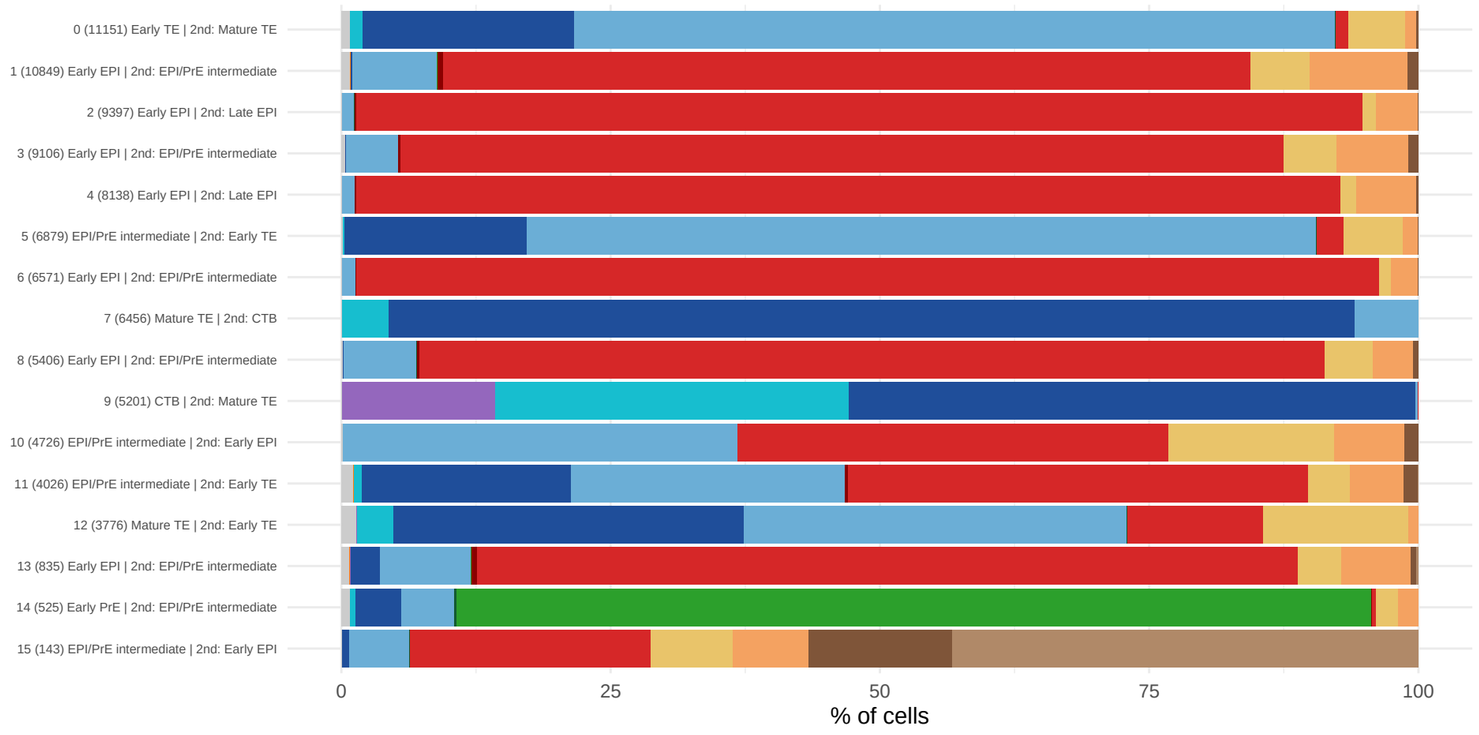


SingleR cluster label



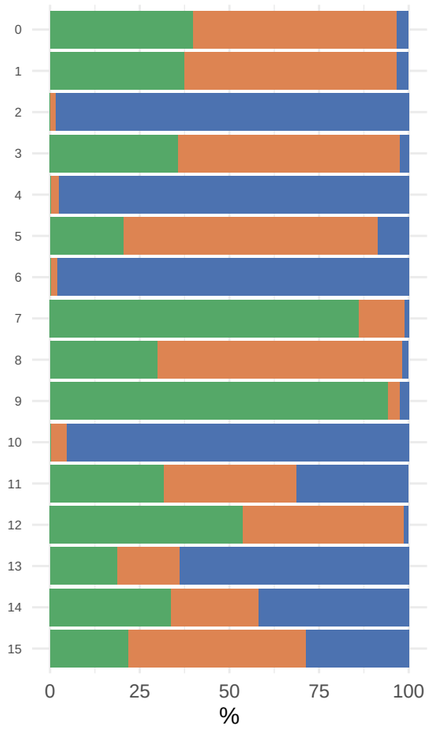
EPI/PrE intermediate
Early EPI
Early PrE
Early TE
Mature TE
CTB

Per−cell labels within each cluster (axis: cluster label, runner−up)



Cleavage
Morula
Prelineage
ICM
EPI/PrE intermediate
Early EPI
Late EPI
Early PrE
Late PrE
Early TE
Mature TE
CTB
STB
EVT
ExE mesoderm
Amnion
Pruned

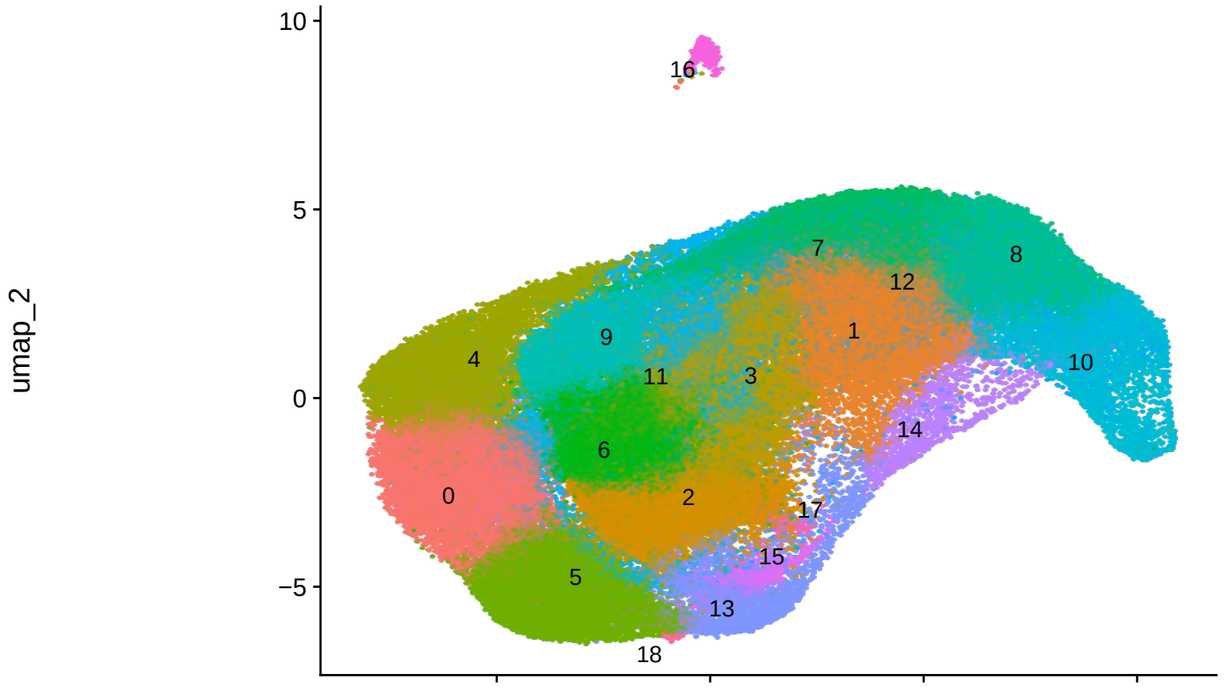
Sample composition



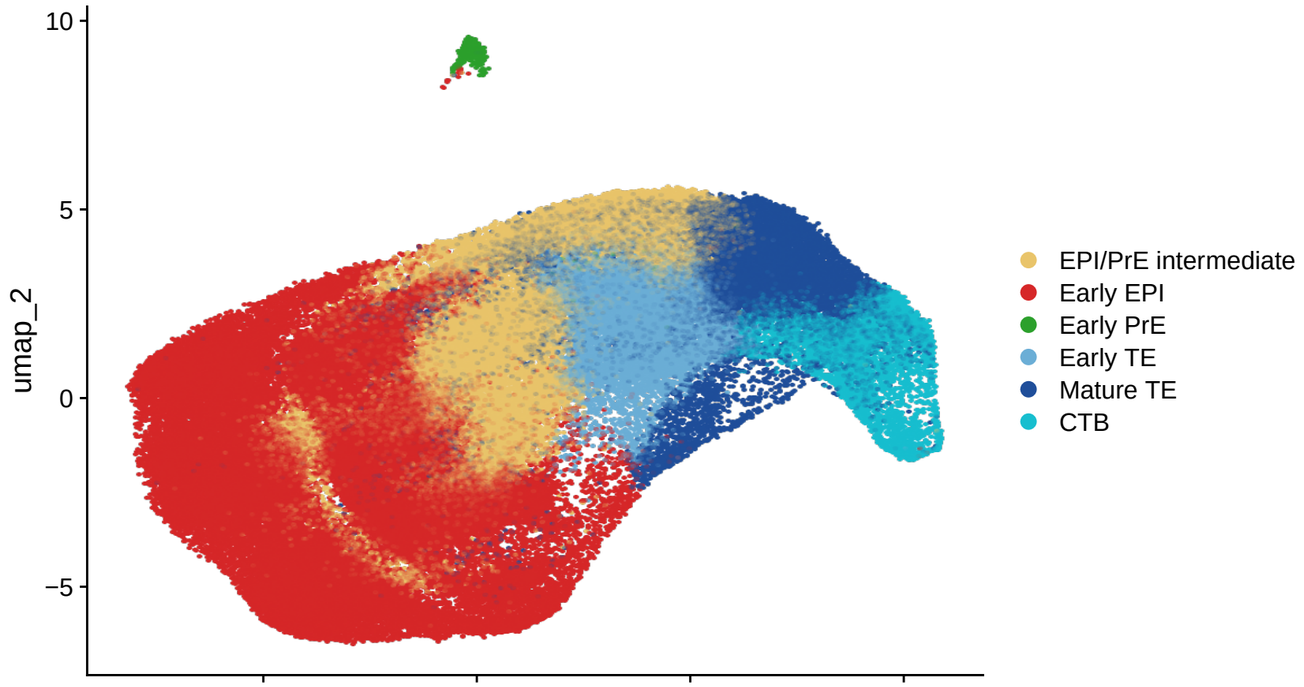
24h−blastoids
48h−blastoids
72h−blastoids

Resolution 0.7

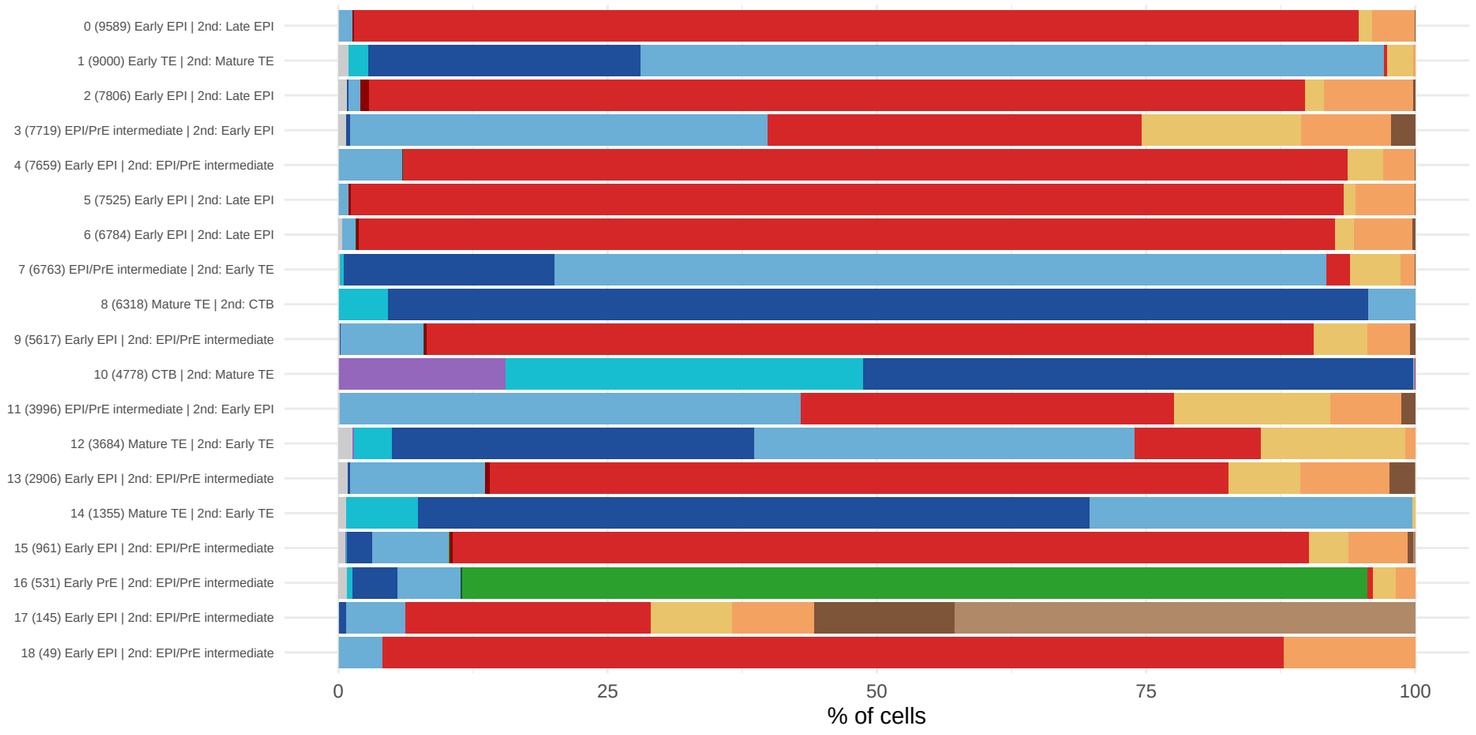
Clusters (19)



SingleR cluster label

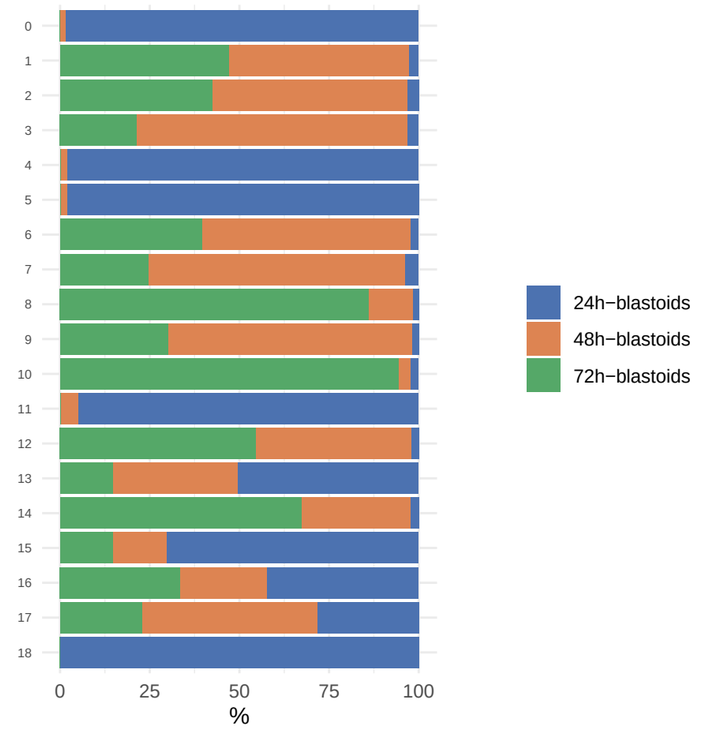


Per−cell labels within each cluster (axis: cluster label, runner−up)



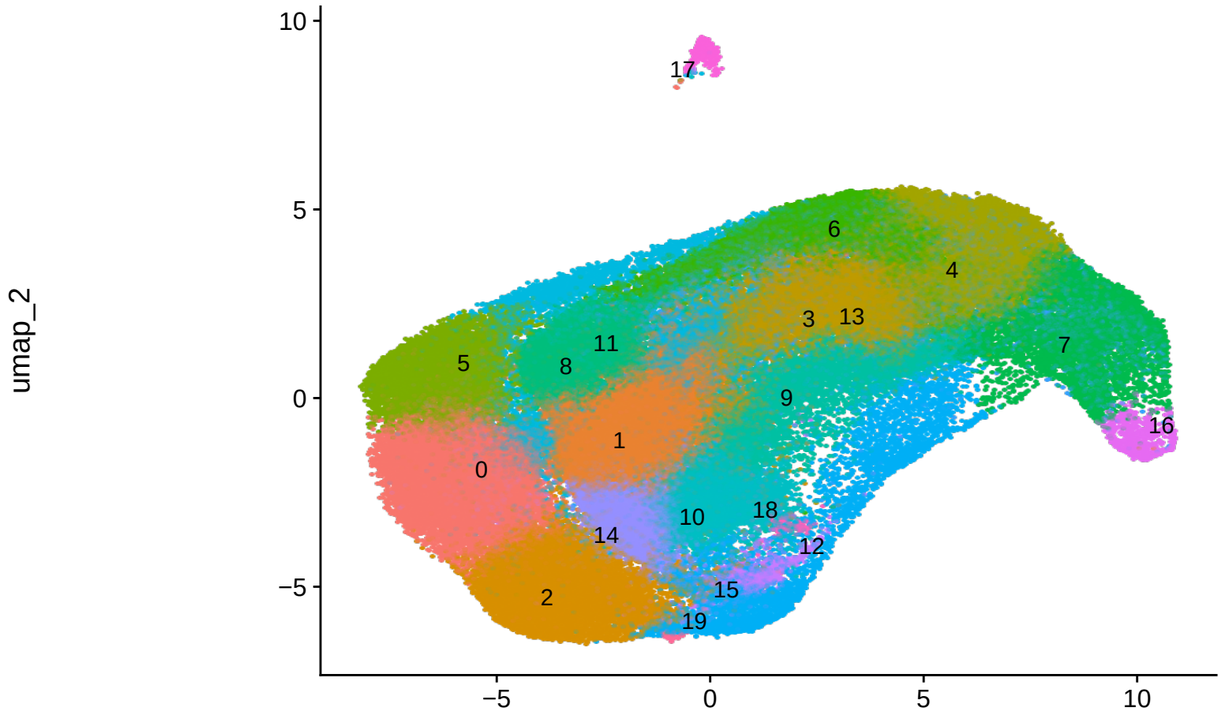


Sample composition

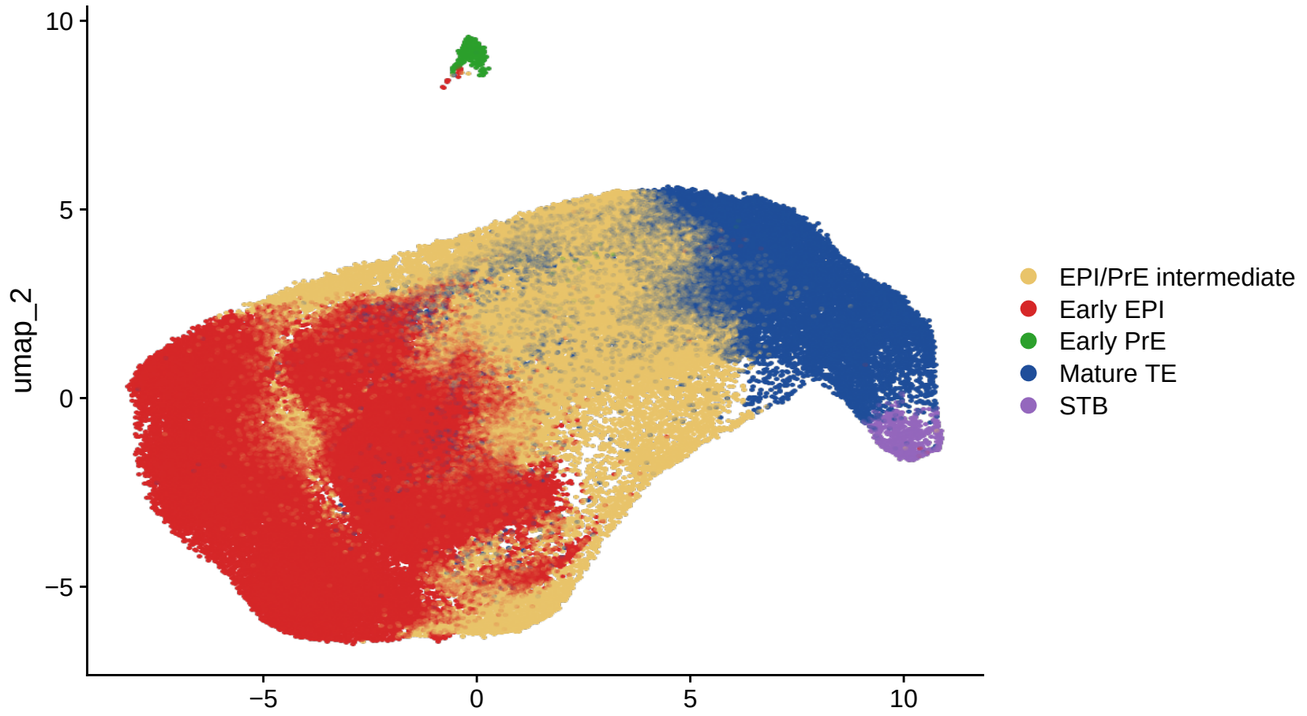


Resolution 0.8

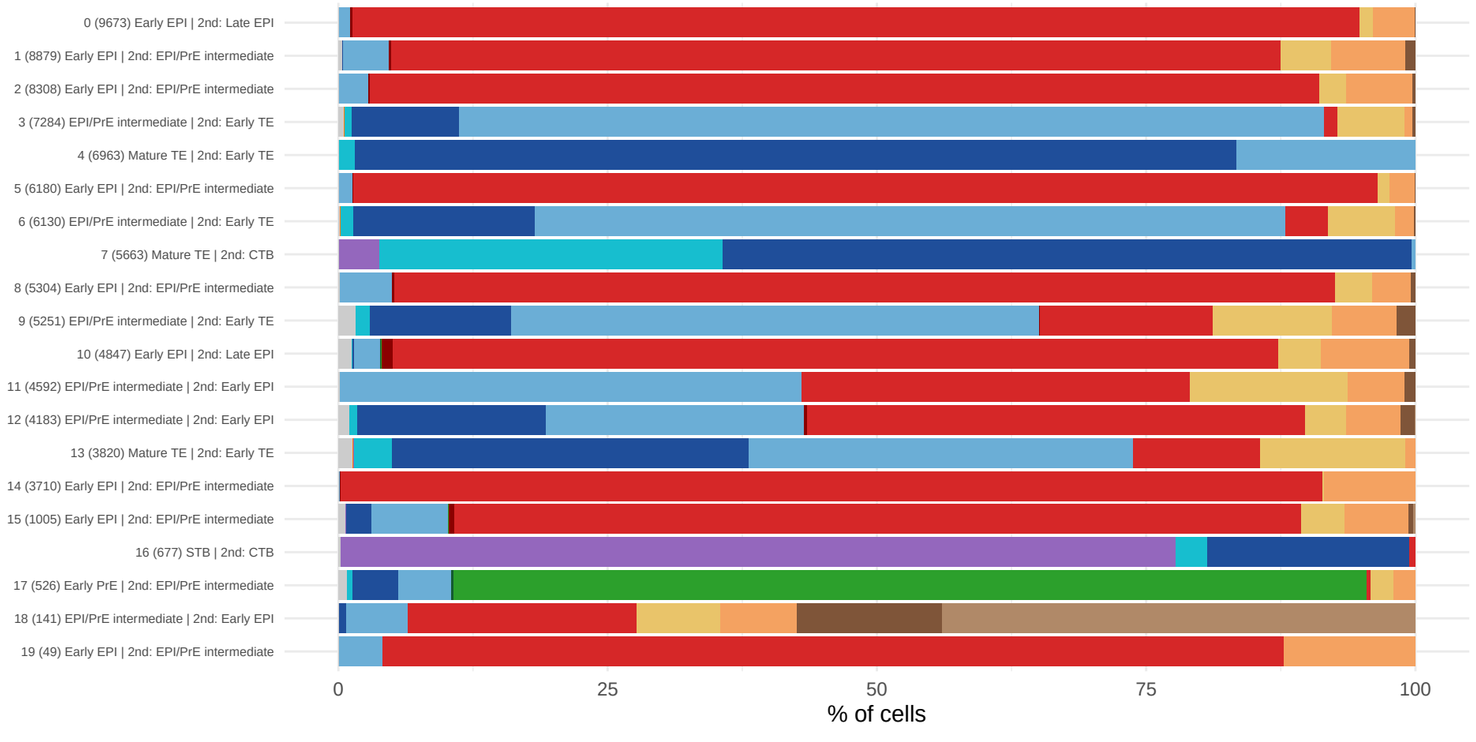
Clusters (20)

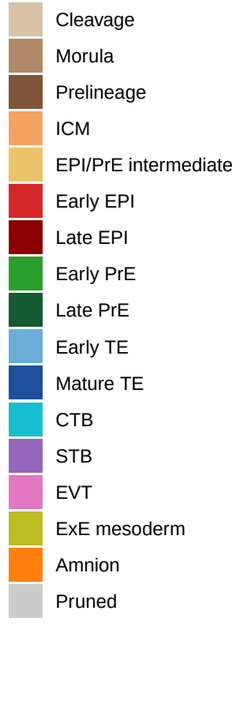


SingleR cluster label

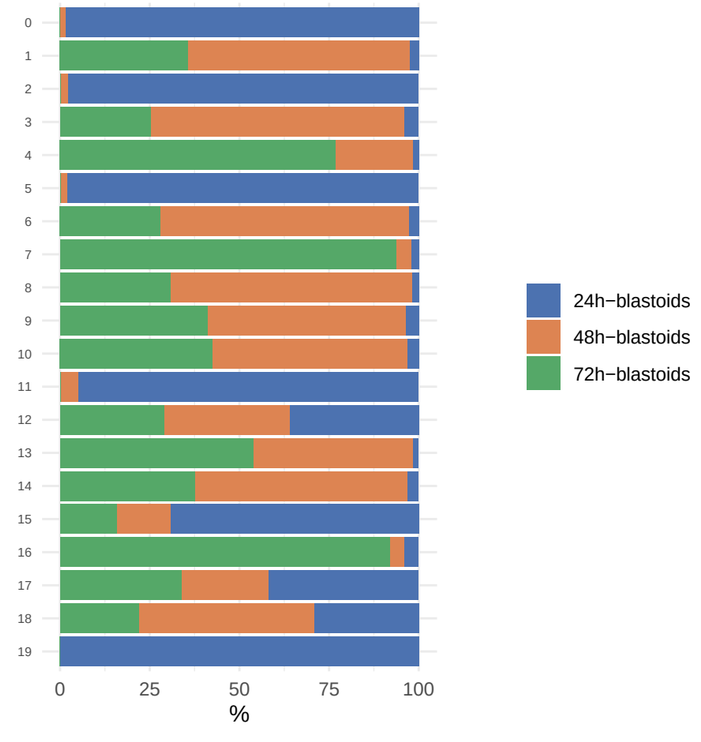


Per−cell labels within each cluster (axis: cluster label, runner−up)



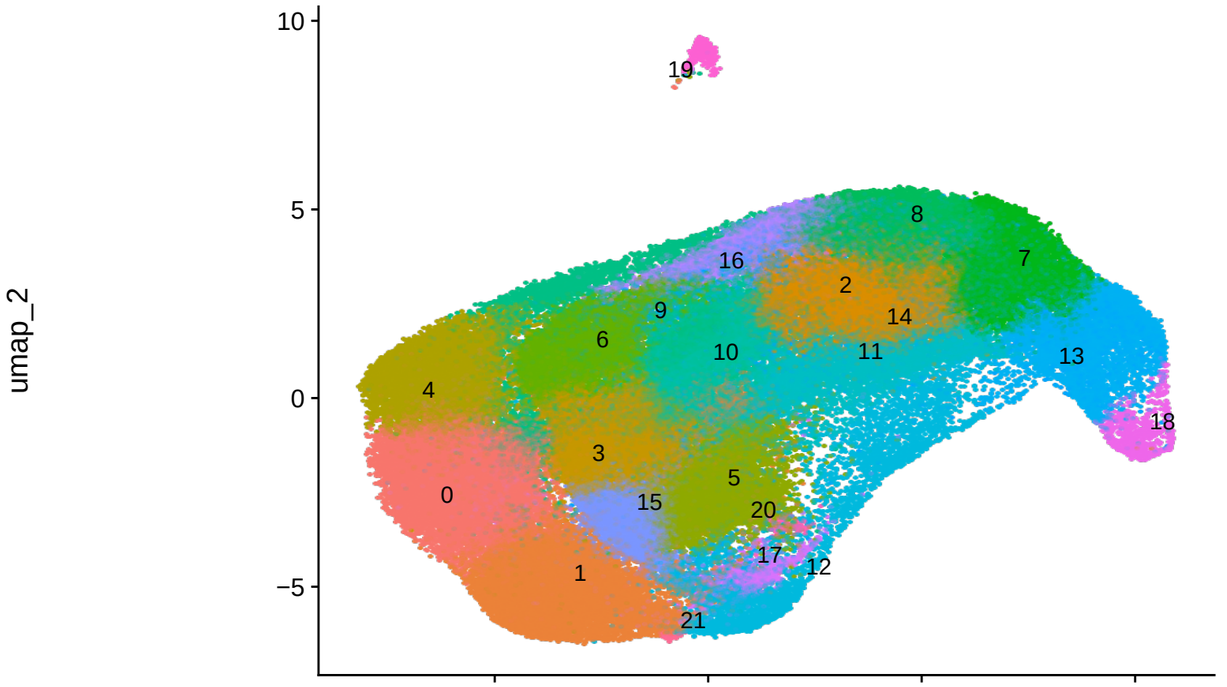


Sample composition

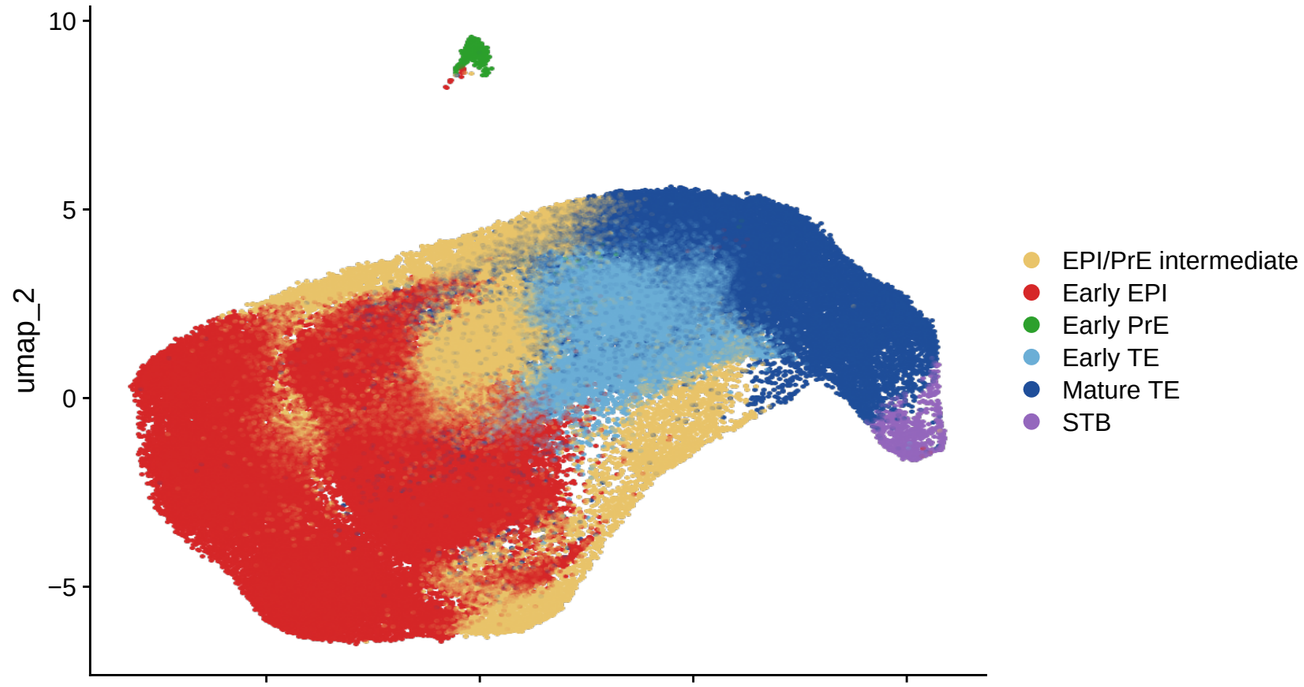


Resolution 0.9

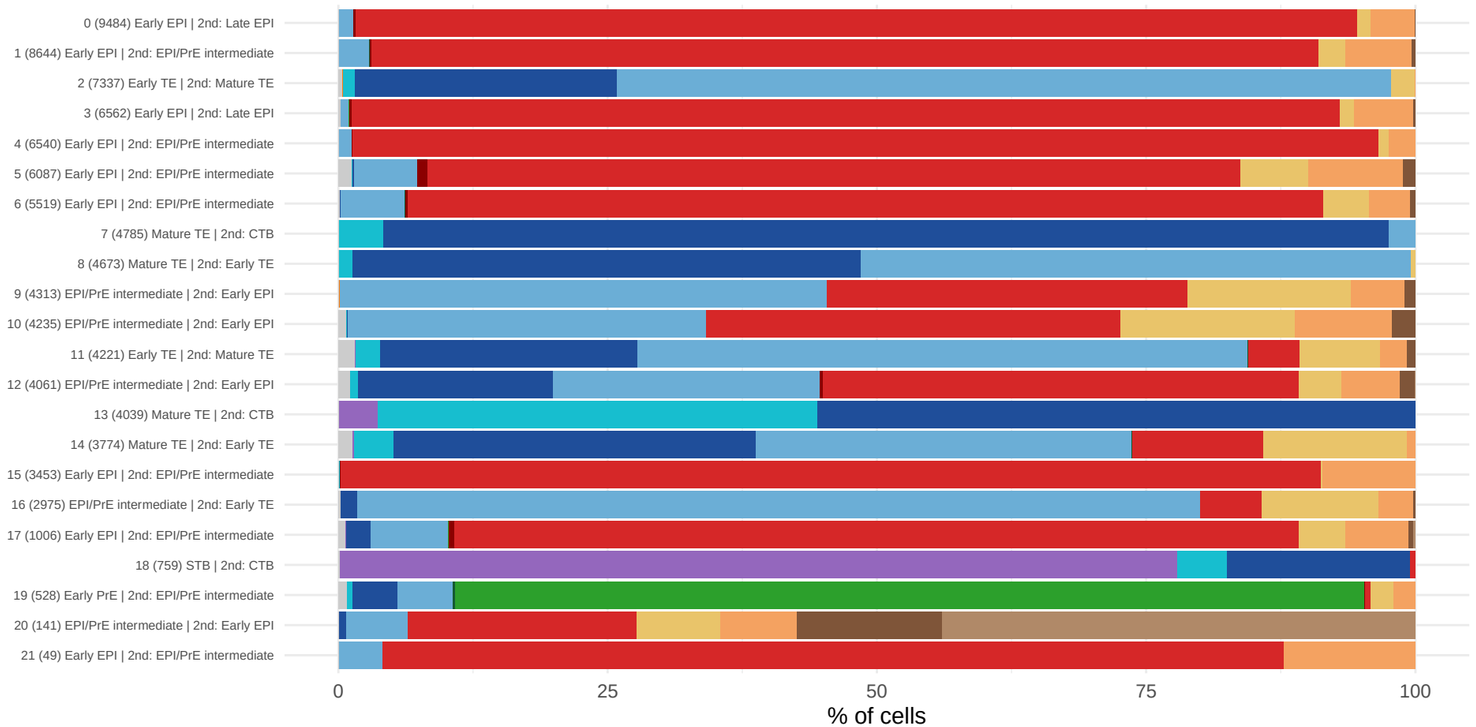
Clusters (22)



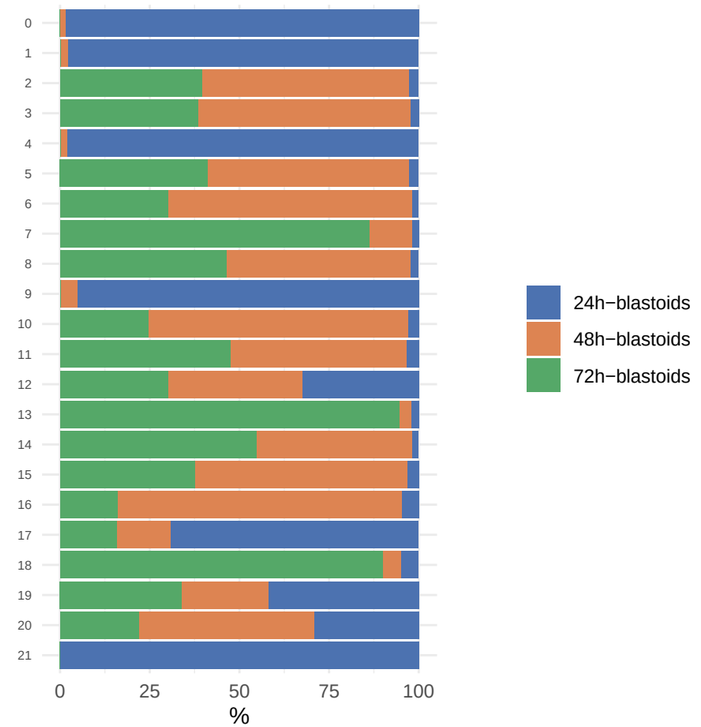
SingleR cluster label



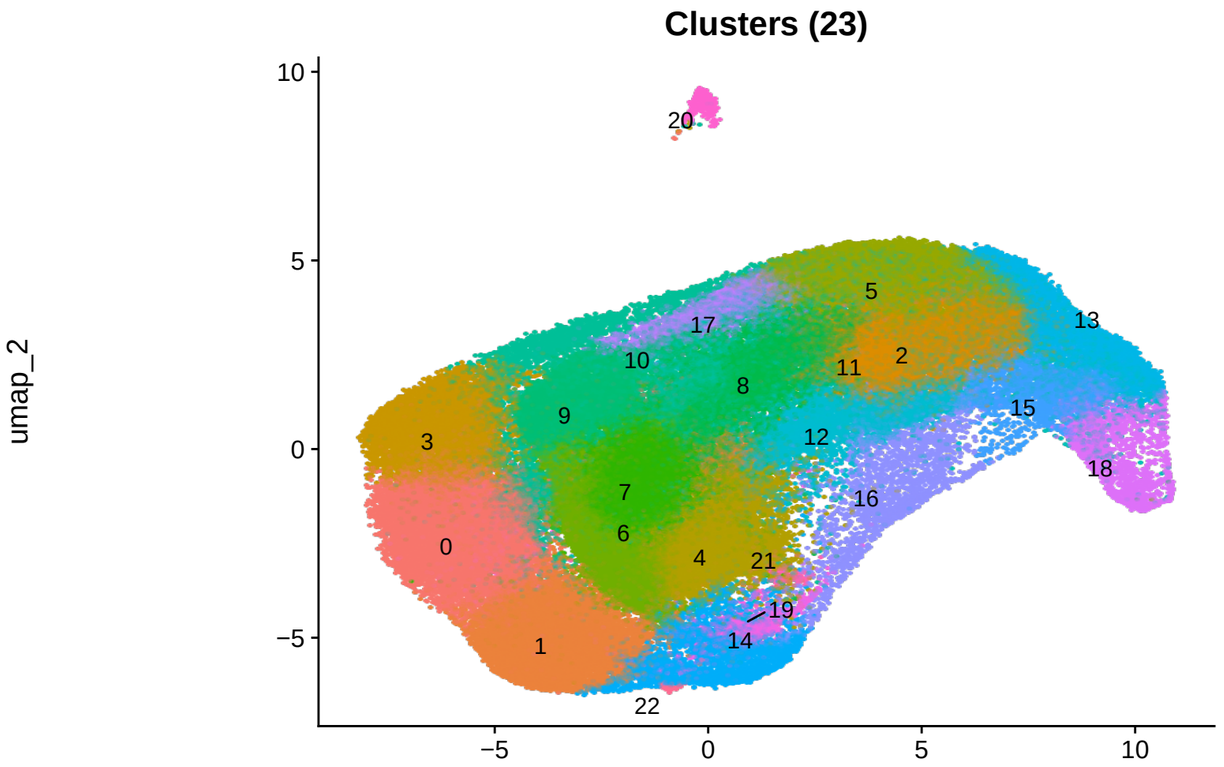
Per−cell labels within each cluster (axis: cluster label, runner−up)

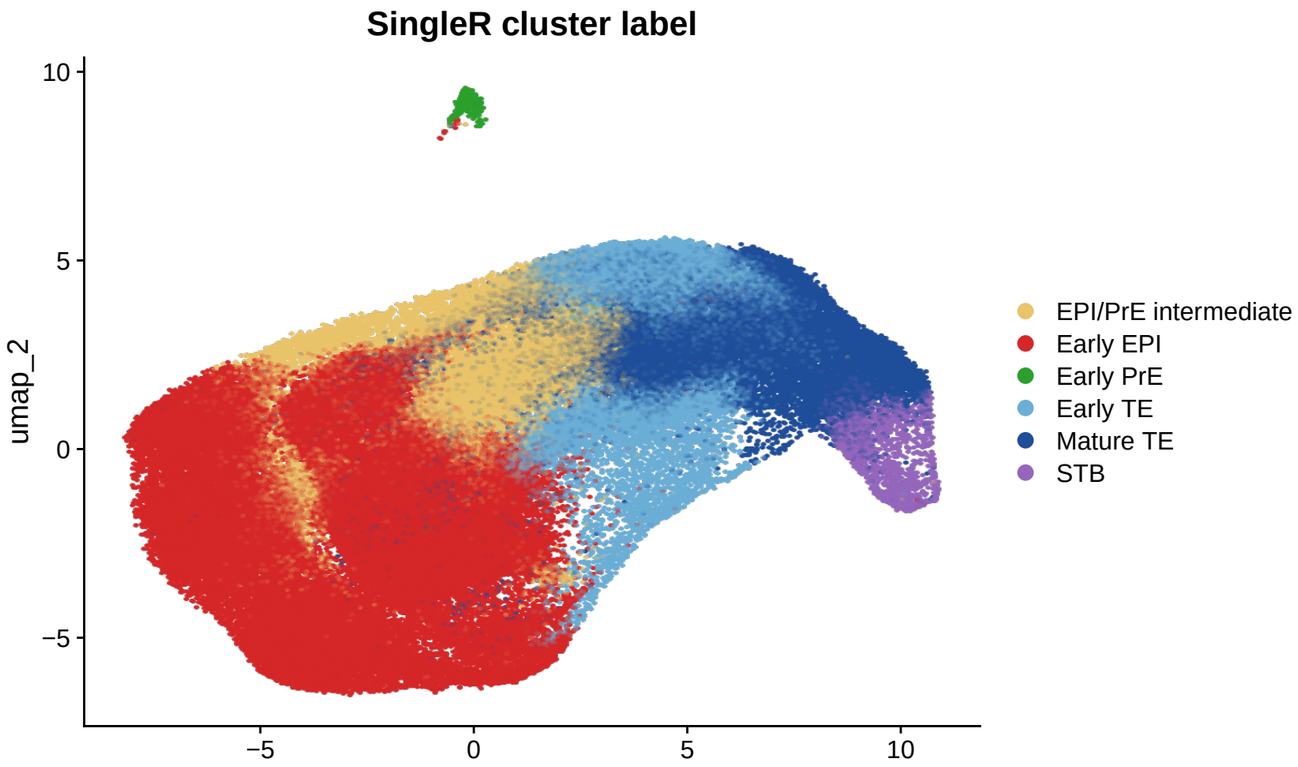


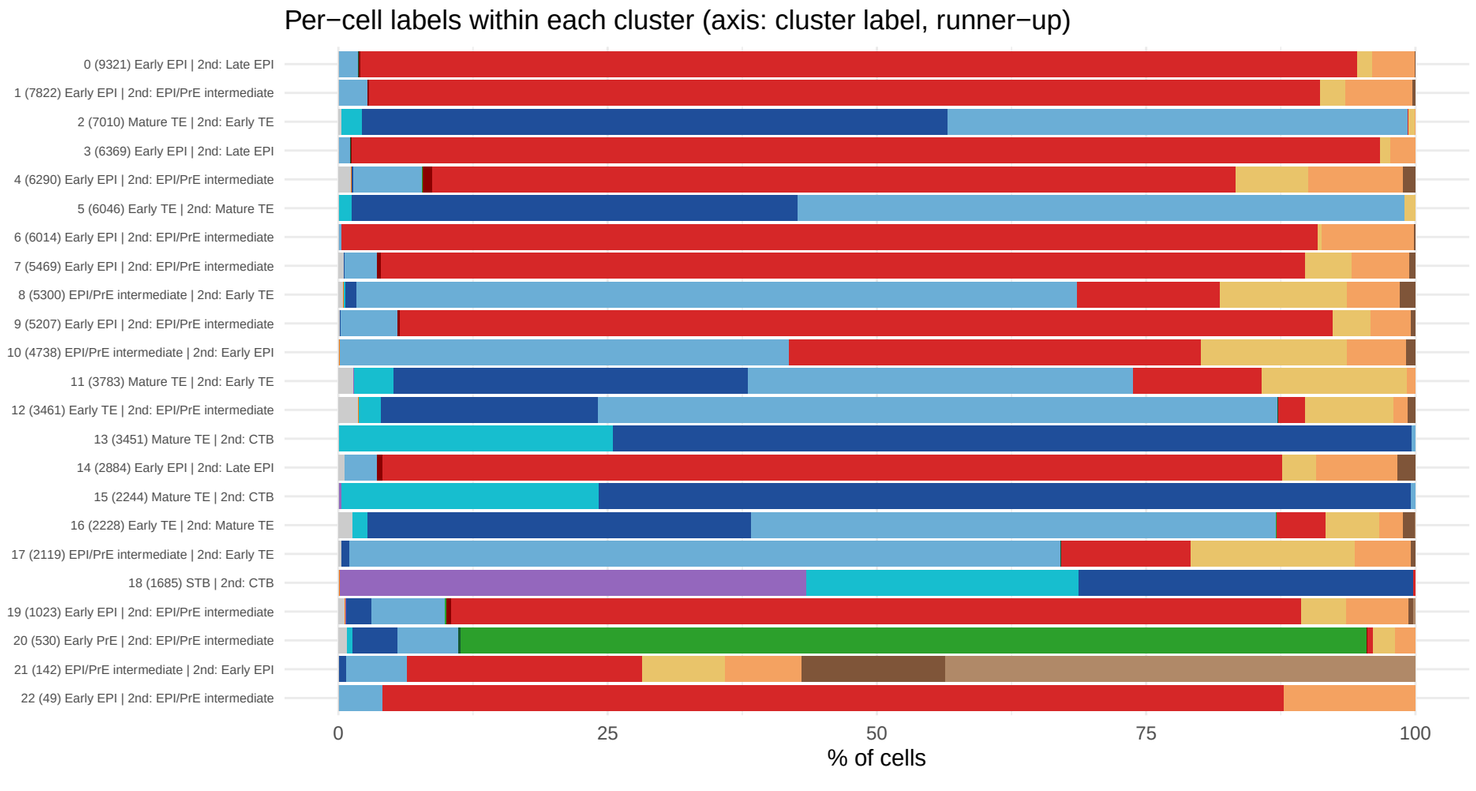
Sample composition

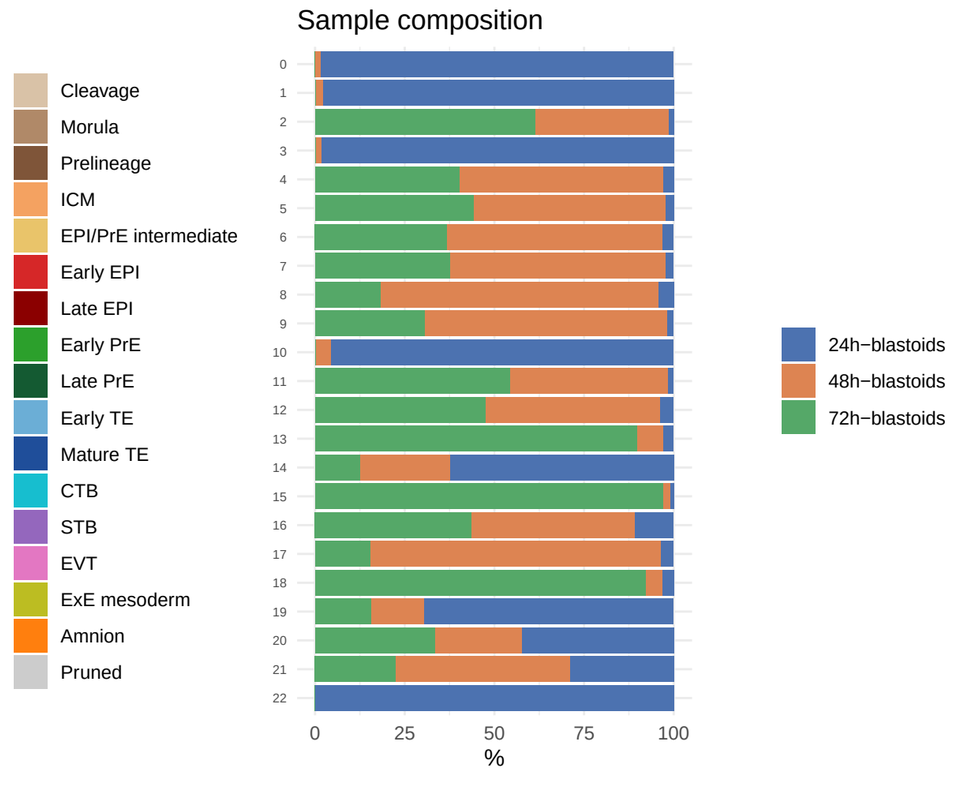


Resolution 1

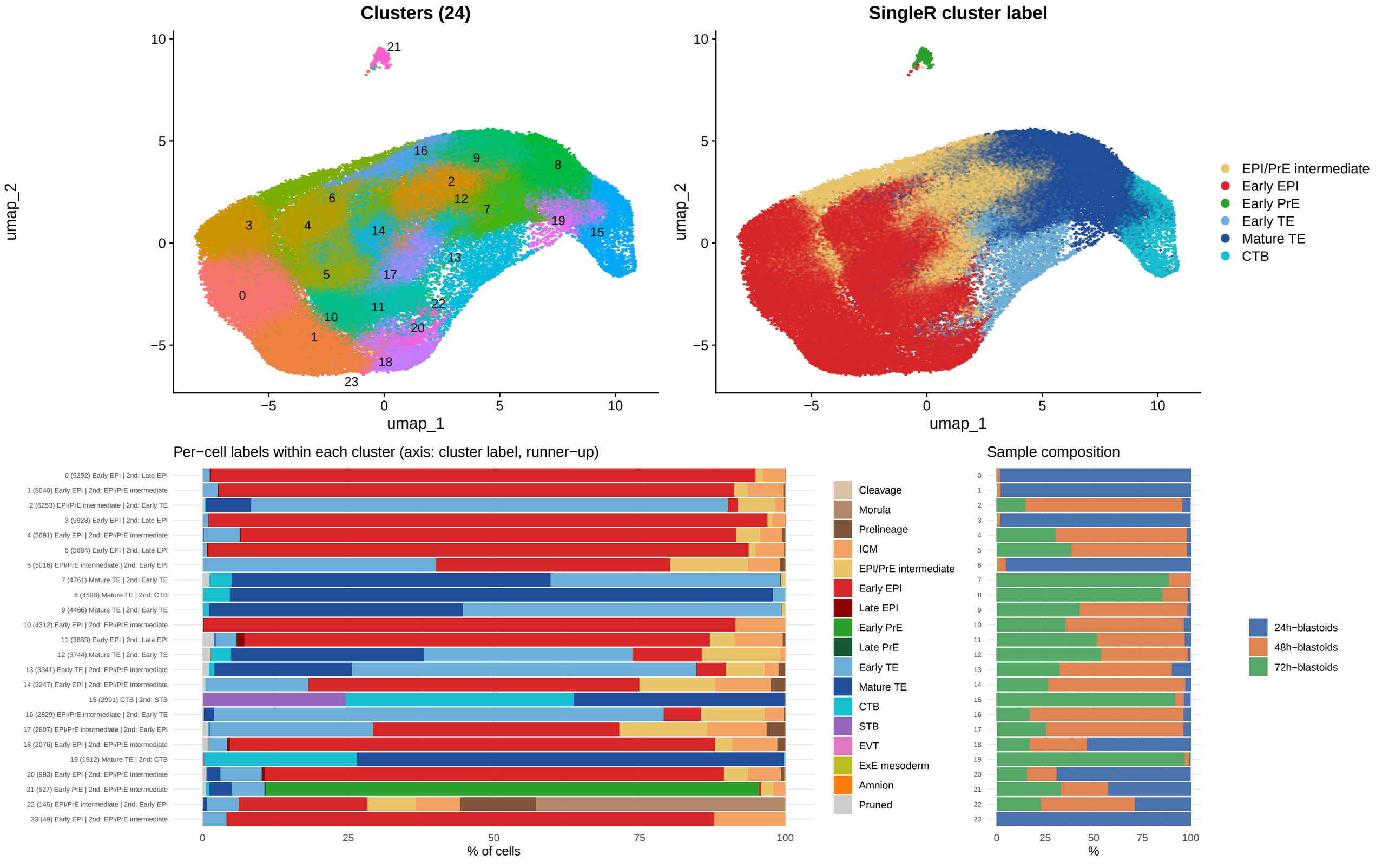




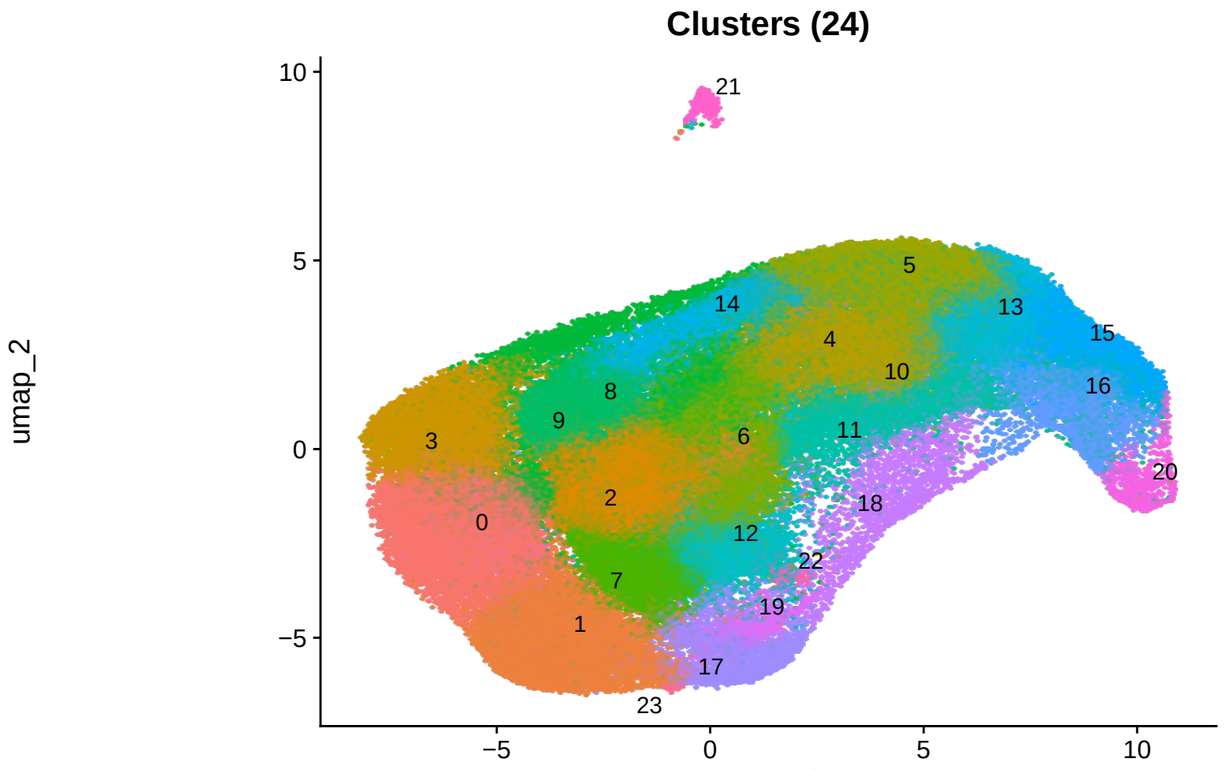


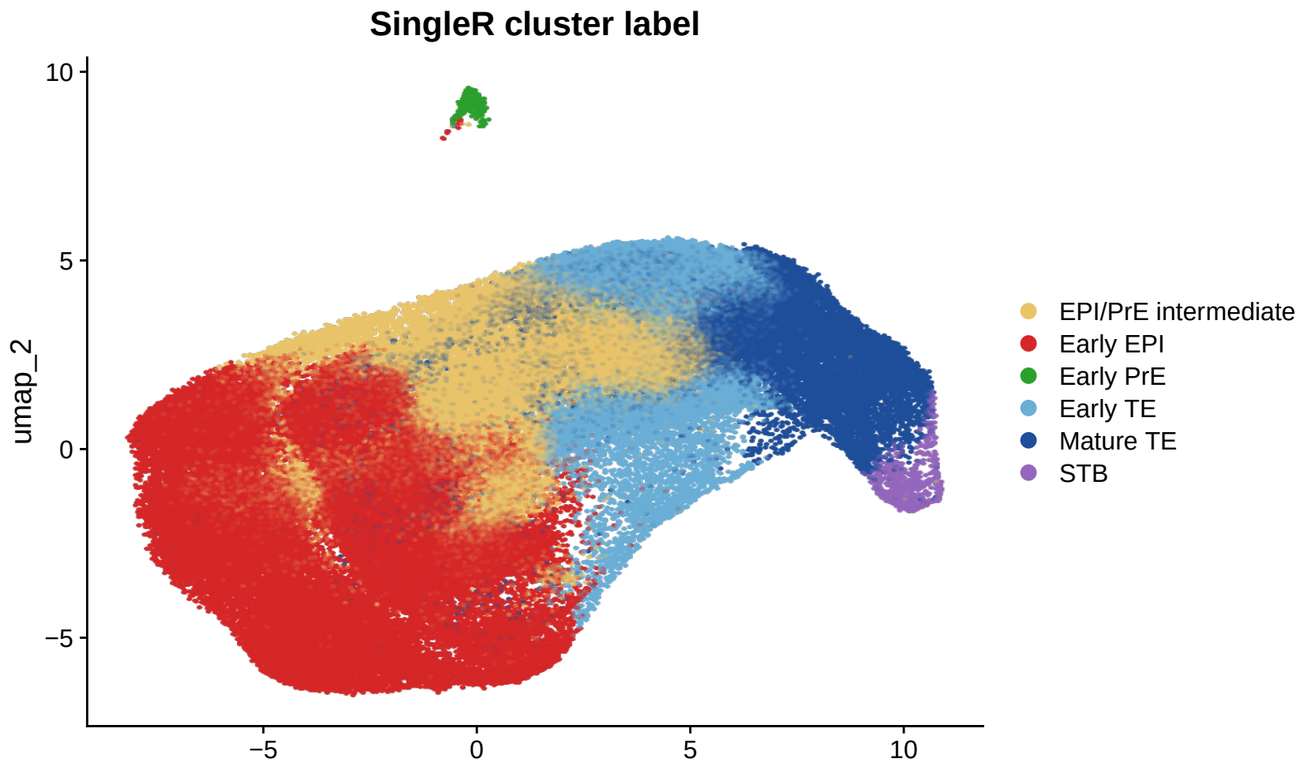


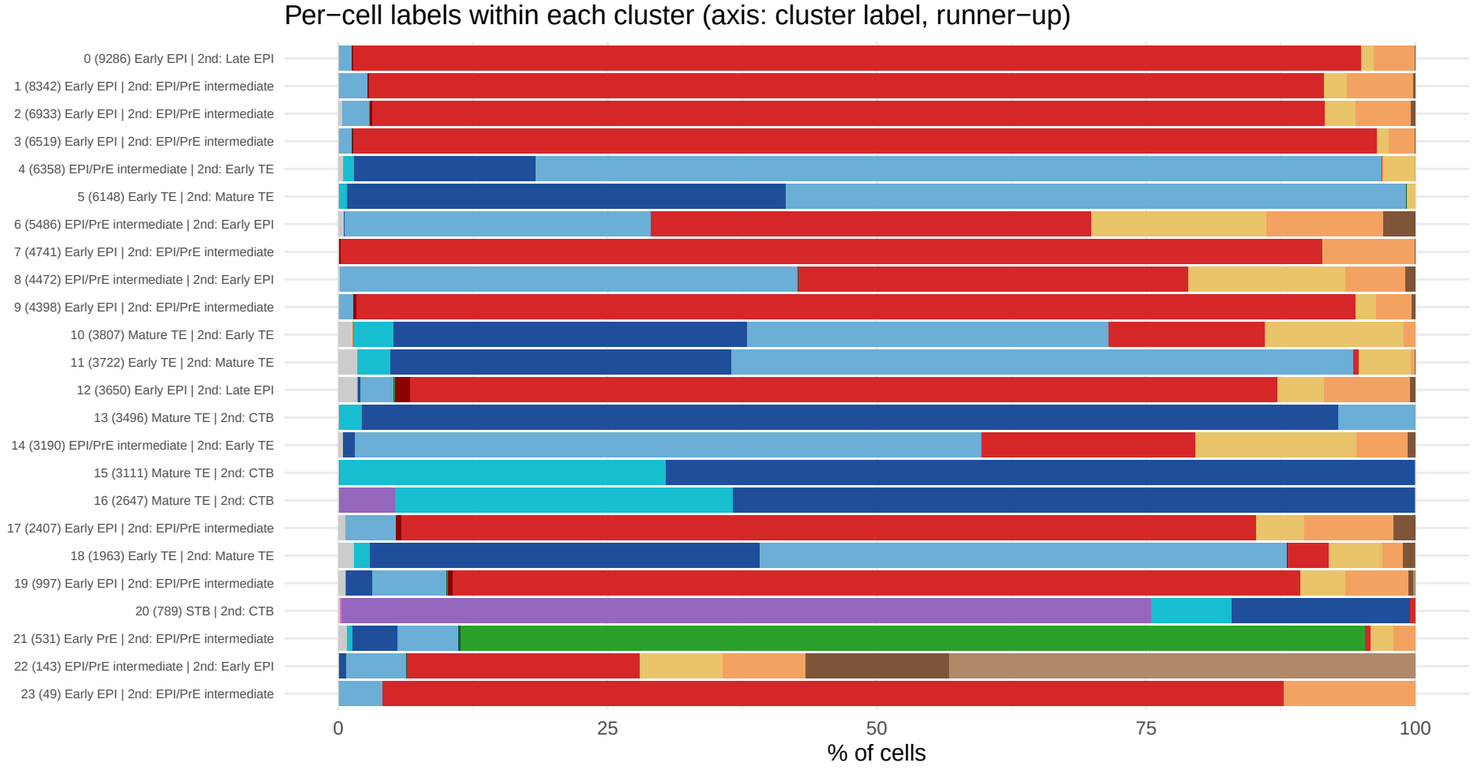
Resolution 1.1

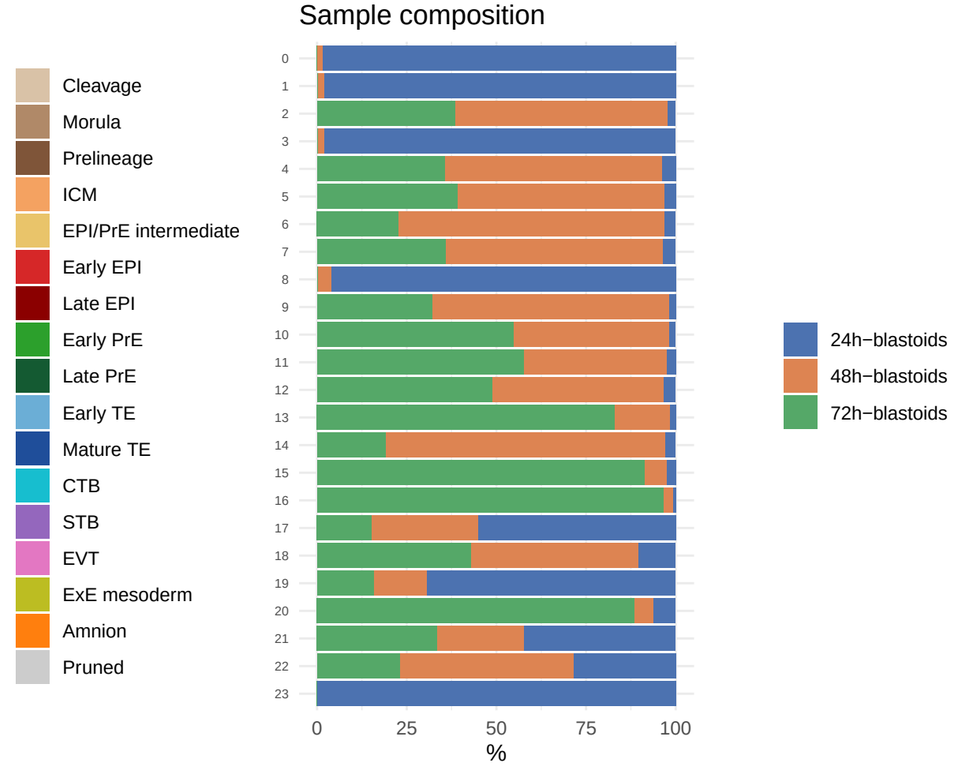


Resolution 1.2



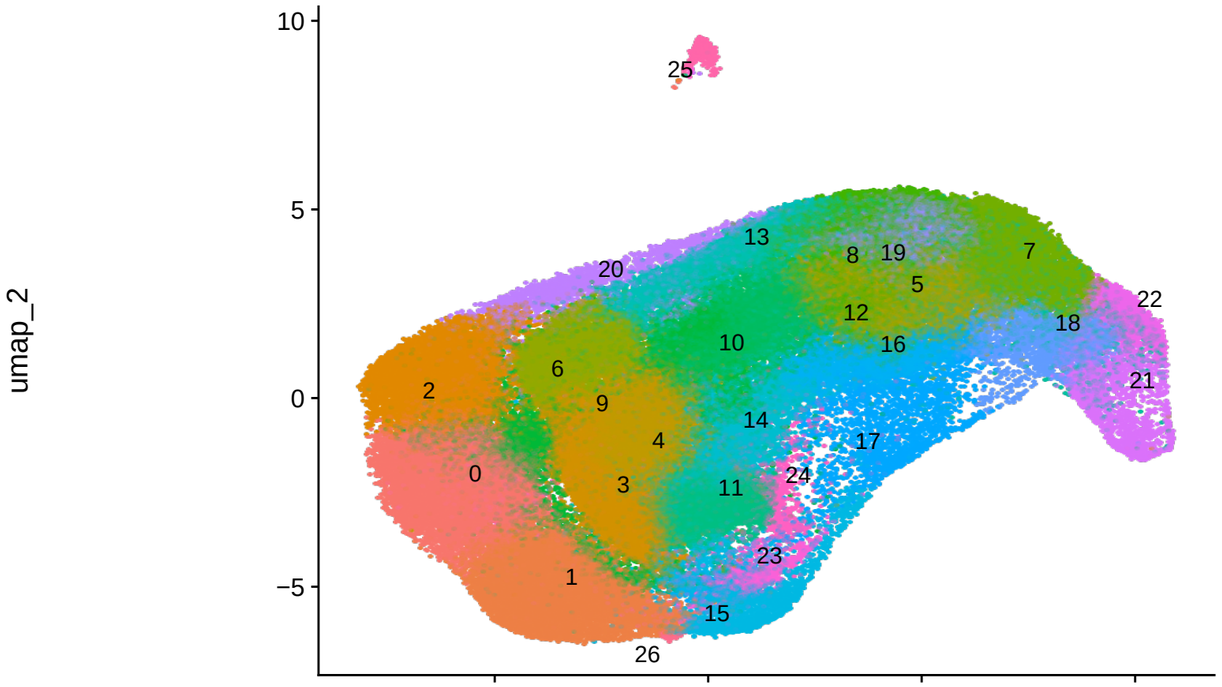




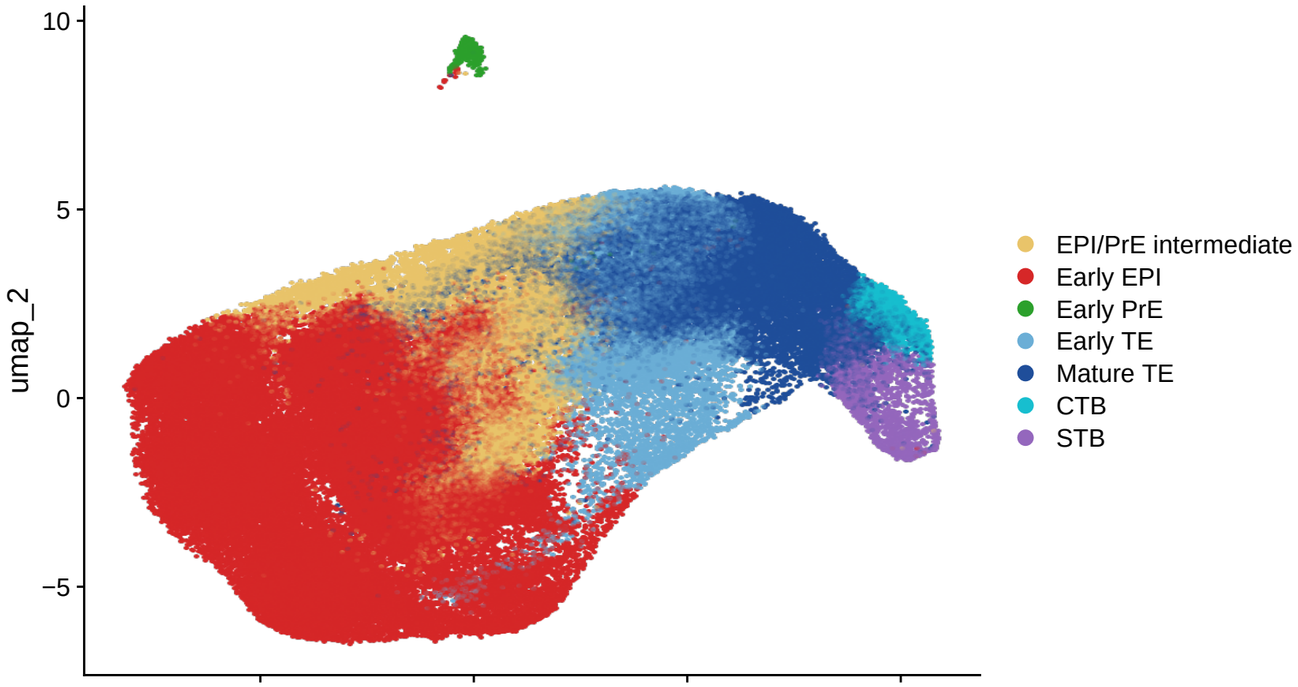


Resolution 1.3

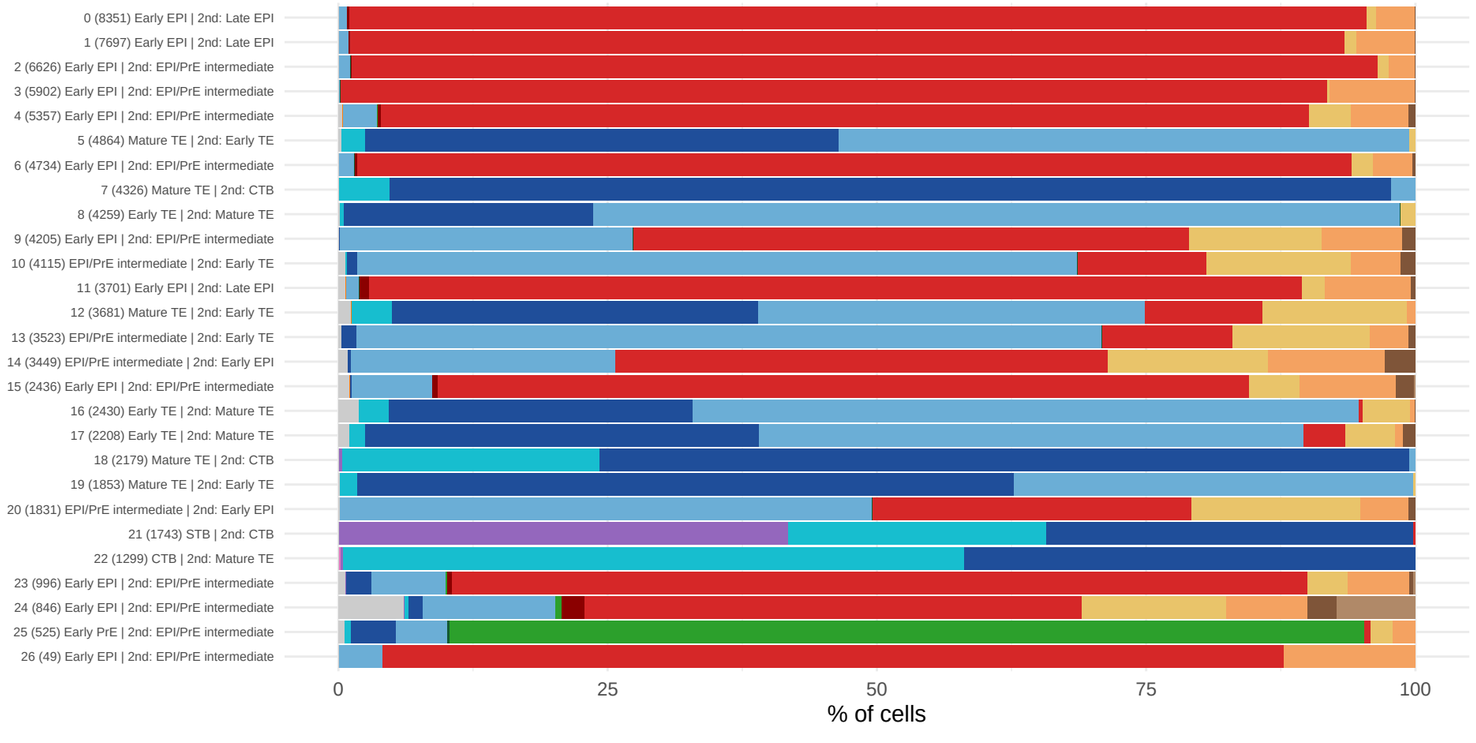
Clusters (27)



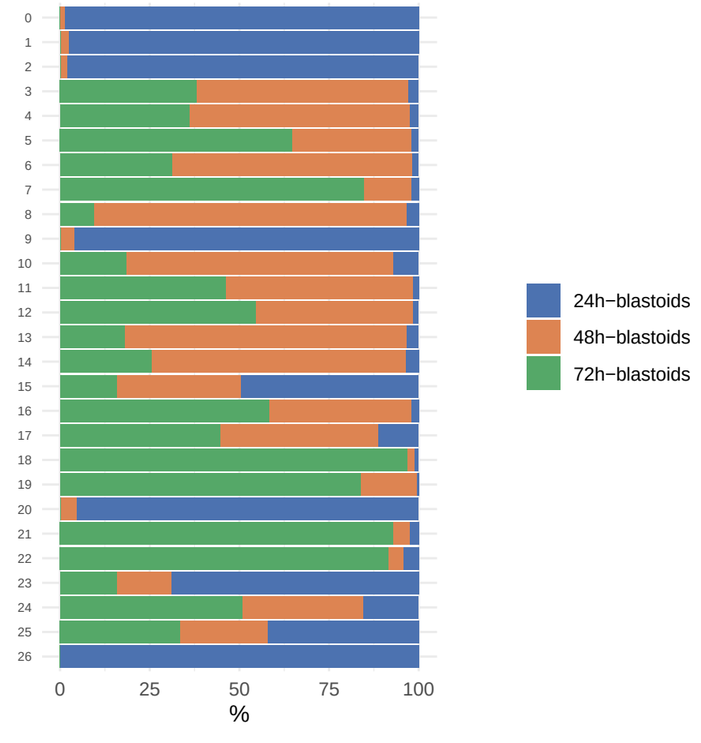
SingleR cluster label



Per−cell labels within each cluster (axis: cluster label, runner−up)

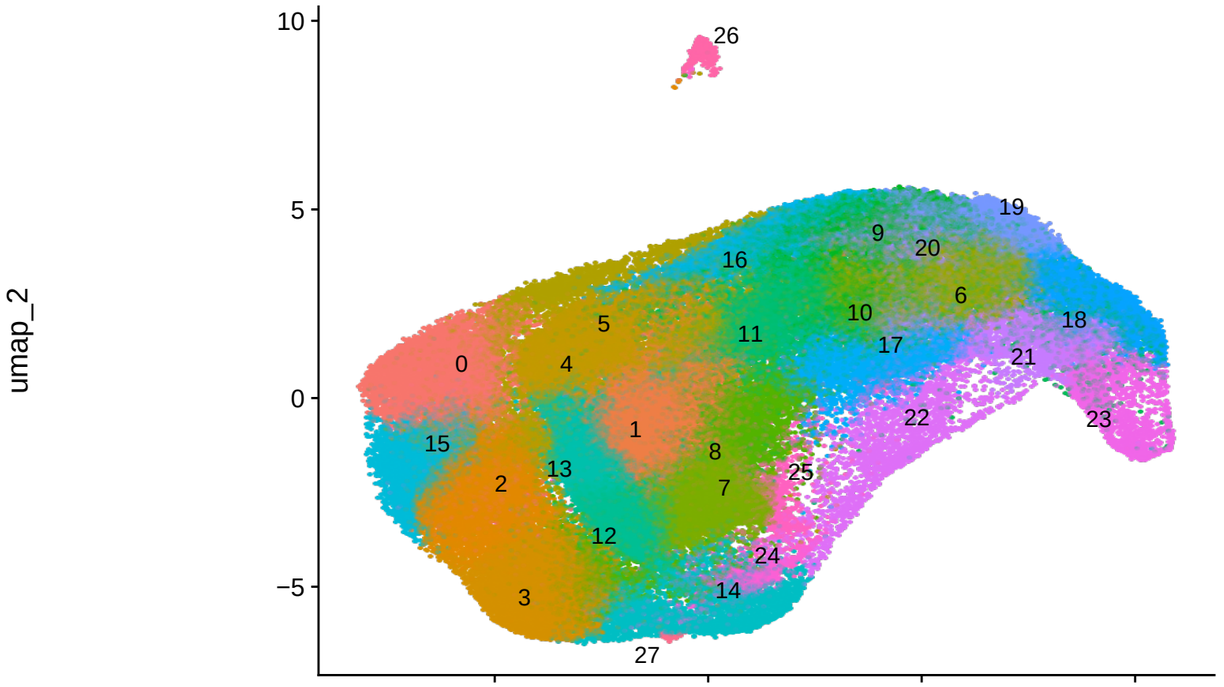


Sample composition

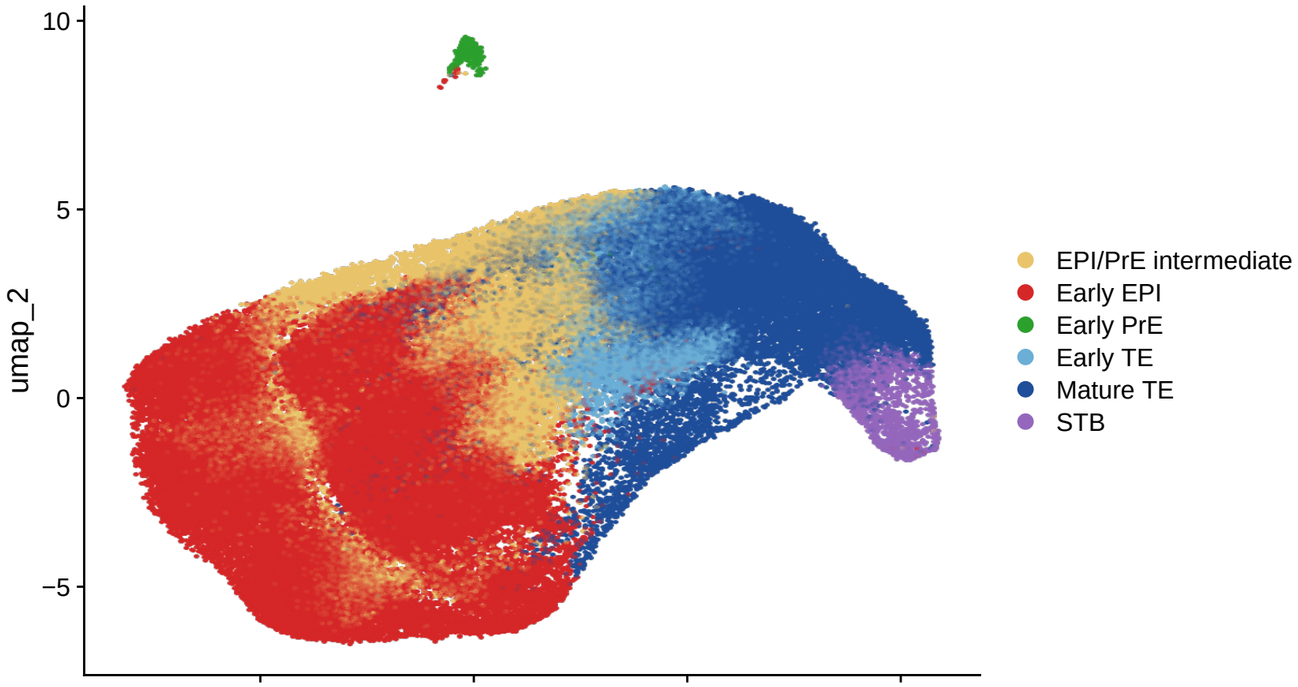


Resolution 1.4

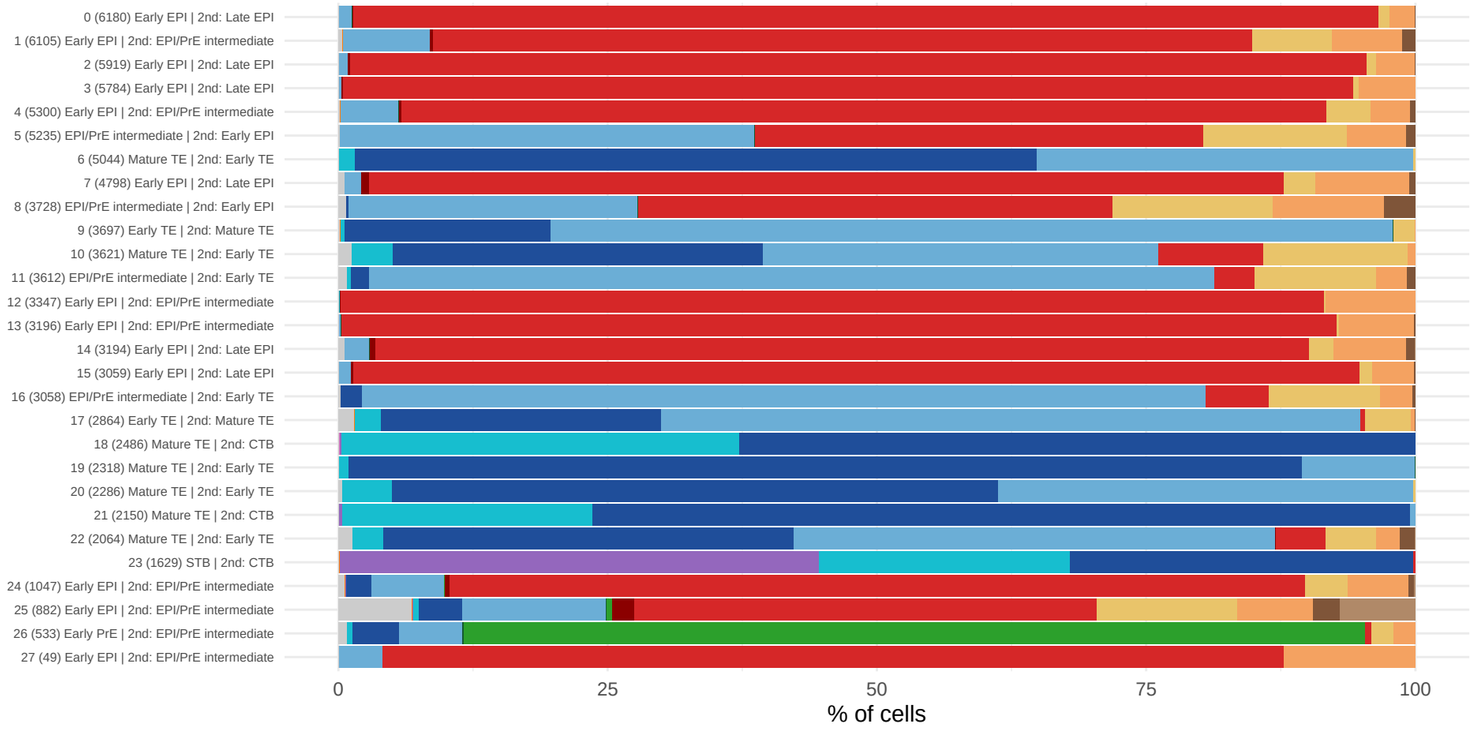
Clusters (28)

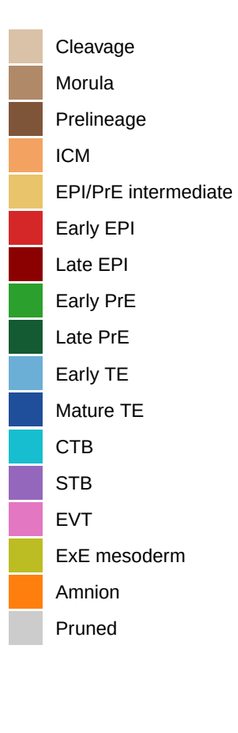


SingleR cluster label

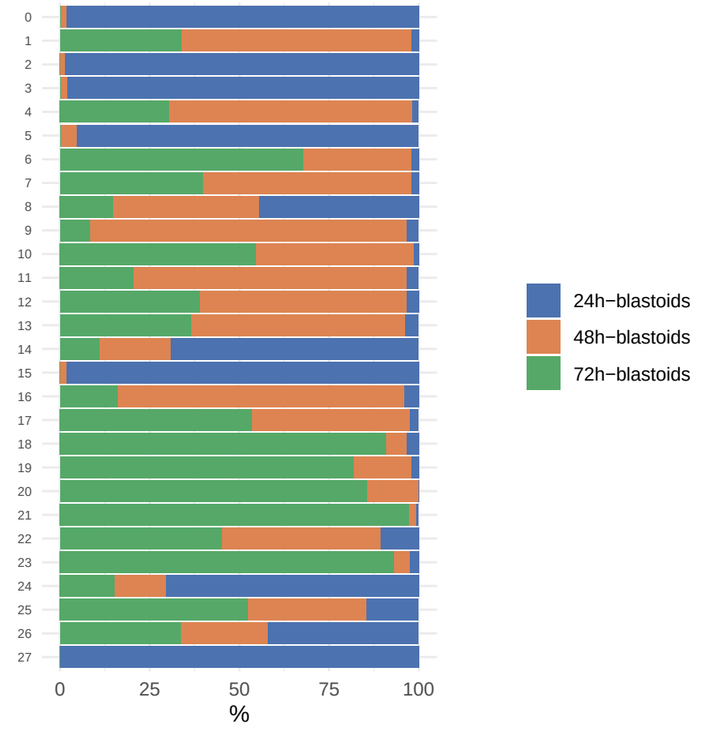


Per−cell labels within each cluster (axis: cluster label, runner−up)



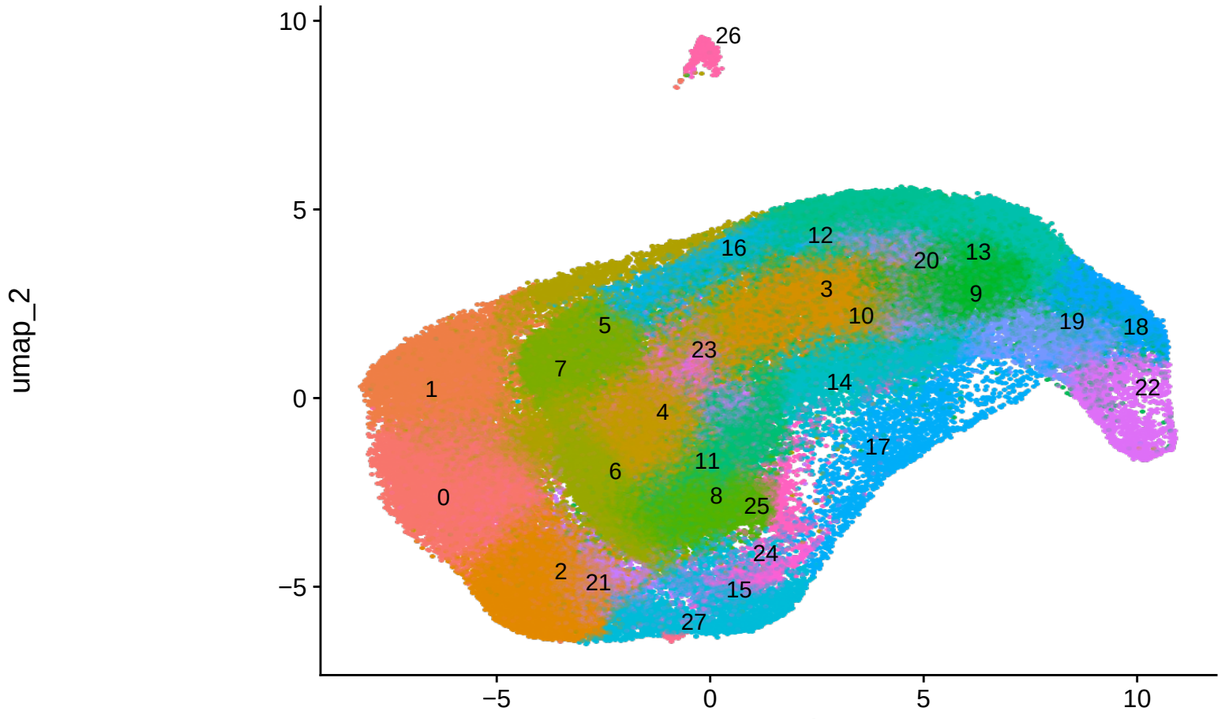


Sample composition

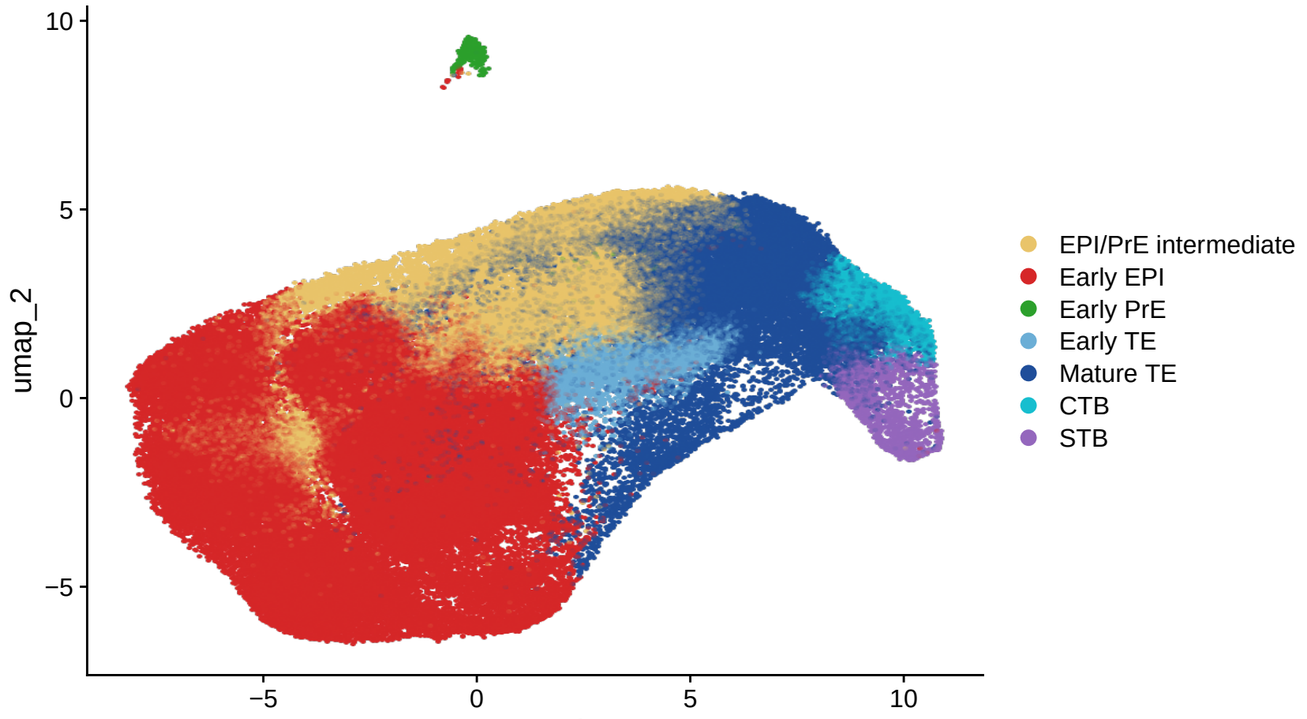


Resolution 1.5

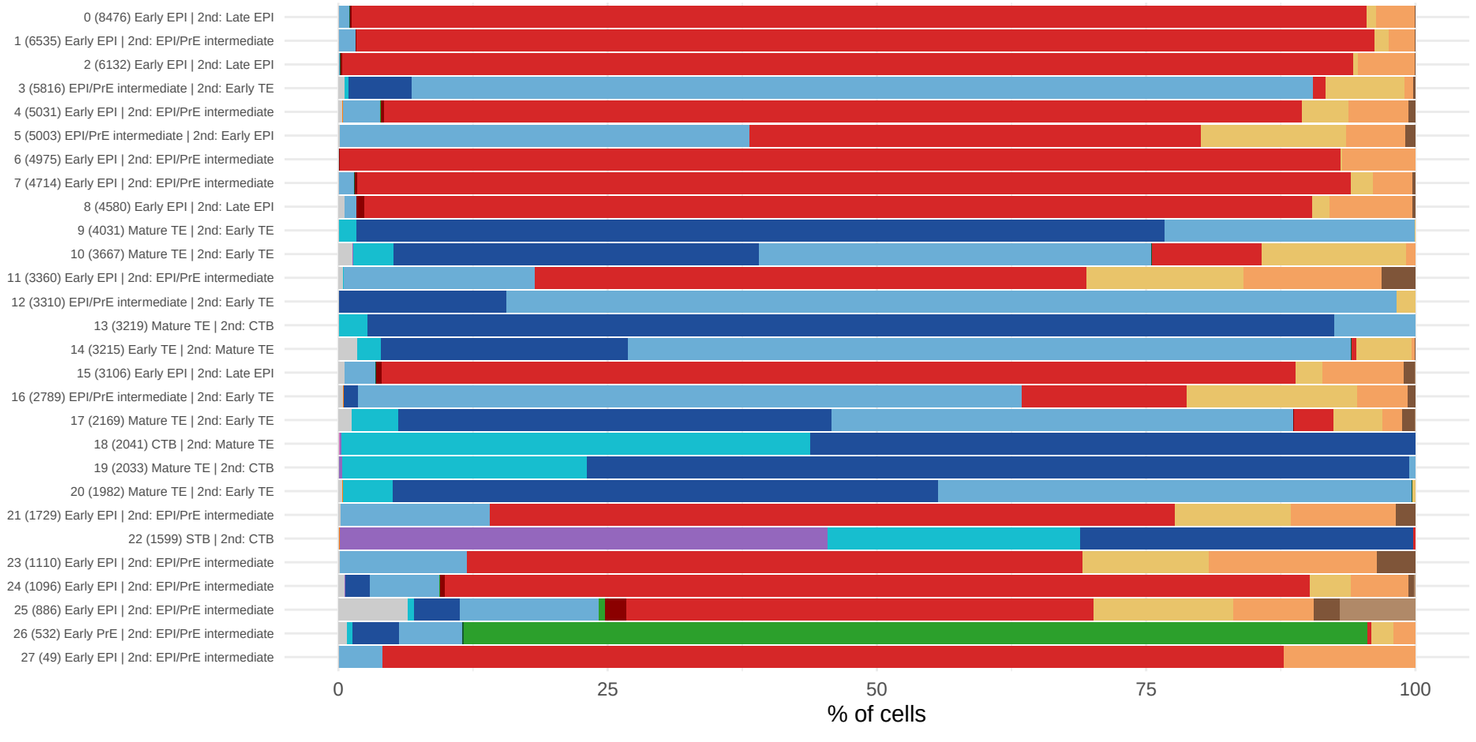
Clusters (28)

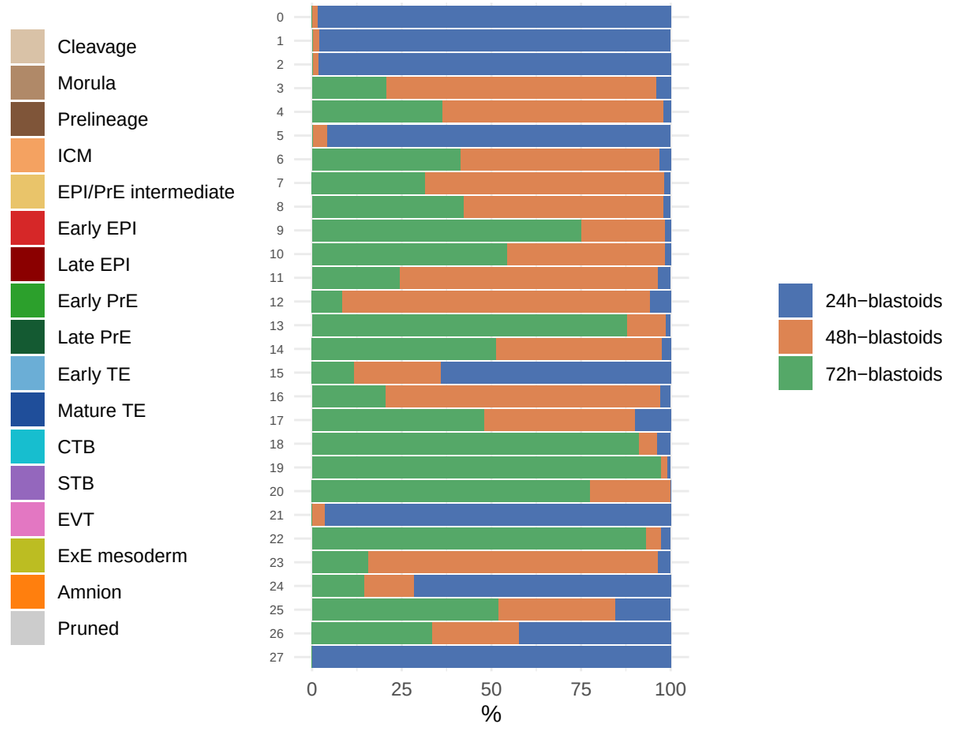


SingleR cluster label



Per−cell labels within each cluster (axis: cluster label, runner−up)





Sample composition

Share of cells per SingleR cluster label at each resolution

Cells

100%
75%
50%
25%
0%

0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
1.1
1.2
1.3
1.4
1.5

Resolution

EPI/PrE intermediate
Early EPI
Early PrE
Early TE
Mature TE
CTB
STB

SingleR cluster label composition by sample at each resolution

resolution: 0.1
resolution: 0.2
resolution: 0.3
resolution: 0.4
resolution: 0.5
resolution: 0.6
resolution: 0.7
resolution: 0.8
resolution: 0.9
resolution: 1
resolution: 1.1
resolution: 1.2
resolution: 1.3
resolution: 1.4
resolution: 1.5

Cells

100%
75%
50%
25%
0%

24h−blastoids
48h−blastoids
72h−blastoids

Cleavage
Morula
Prelineage
ICM
EPI/PrE intermediate
Early EPI
Late EPI
Early PrE
Late PrE
Early TE
Mature TE
CTB
STB
EVT
ExE mesoderm
Amnion